

ANNUAL REPORT

1990-91



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NAARM

National Academy of Agricultural Research Management



ANNUAL REPORT

1990-91



NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT
Rajendranagar, Hyderabad-500 030 (India)

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CONTENTS

Introduction	1
Objectives	2
Highlights	3
Training Programmes	5
Course for Junior Scientists	6
Course for Middle and Senior Scientists	12
Workshops	21
International Workshop on Management Information Systems for Agricultural Research	24
Consultancy Services	33
Organisational Structure and Salient Activities	34
Agricultural Research Systems and Policies Unit	35
Educational Systems and Technology Unit	42
Transfer of Technology Systems and Policies Unit	45
Project Management Unit	48
Human Resources Development Unit	51
Information and Documentation Unit	54
Centralised Services	58
Human Resources	62
Financial Outlay	63
Physical Facilities	64
Distinguished Visitors	66
Publications	70
Miscellanea	71
Faculty Development	72
NAARM Alumni Association	76
Important Committees	77
Personnel	82

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INTRODUCTION

Historical Background

It is now well recognised that management holds the key to increased productivity in research, as it does to production and productivity in several other sectors. Management principles are being increasingly applied in agricultural research, education and extension. The need for an institutional set-up to provide training on principles of scientific management of agricultural research was felt two decades ago by the Indian Council of Agricultural Research (ICAR) and the Government of India, following the organisation of a training course in Agricultural Project Management, in response to a request from the FAO. A National Training Centre for Agriculture Development and Project Planning functioned at Indian Agricultural Research Institute (IARI), New Delhi, during the Fourth Five Year Plan. This further strengthened the need to continue these training activities, and to establish an organisation for this purpose.

A working group, set up during the Fifth Five Year Plan, recommended the establishment of a Central Staff College for Agriculture to function as national and international repository of ideas and information in the field of agricultural research and development. The proposed college was also required to act as a clearing house for dissemination and utilisation of Research Management information and also to undertake a programme of publishing high quality training material and allied publications considered important for the priority areas of education.

After more detailed deliberations and planning, the Central Staff College for Agriculture came into existence on September 1, 1976. This was later renamed as the National Academy of Agricultural Research Management.

The Academy is located at Rajendranagar, Hyderabad, at the campus of the Andhra Pradesh Agricultural University (APAU) on 50 hectares of land provided by the University to the Academy on a long-term lease. The NAARM campus is situated at Rajendranagar, a suburb of Hyderabad, about 15 km south of the city.

Several functional buildings have now been constructed in this campus. These include an Administrative Block, Faculty-cum-Laboratory Building, an Academic Block which has several class and discussion rooms, Halls of Residence and Scientists Home. Besides, residential quarters for all categories of staff have also been constructed. Several instructional and technology demonstration facilities have been developed. An Advanced Centre on Agricultural Education and Research Management has also been established with the support of UNDP/UNESCO/FAO. These physical and infrastructural facilities as well as the technical strength of the programmes have placed the Academy as an important National and International Centre serving in the country and the South and South East Asian Region.

OBJECTIVES

The National Academy of Agricultural Research Management (NAARM) was established with the main objective of familiarising all those concerned with the agricultural research, education and development with the principles of scientific management. It had taken up the responsibility to plan, organise and conduct training programmes, workshops and seminars in various areas related to agricultural research management for the research scientists of ICAR and agricultural universities. It is functioning as a repository of ideas and information both at national and international levels, and fulfilling its role as a clearing house for dissemination and utilisation of such information and know-how. The Academy had also started training programmes in the areas of educational technology, curriculum development and structured learning. Research programmes and case studies in various fields of agricultural research management and educational technology were also taken up.

The main objectives of the Academy are :

- * to organise and conduct training programmes in agricultural research management to scientists at various levels,
- * to undertake systematic review and study of management problems of ICAR research institutes, agricultural universities, agricultural research projects and systems,
- * to organise, liaise and coordinate programmes of international cooperation in the field of agricultural research management,
- * to plan, organise and conduct training programmes, workshops and seminars in educational technology and structured learning for the faculty of agricultural universities and ICAR institutes and
- * to assist the State Agricultural Universities (SAUs) in developing regional centres of management to cater to the needs of the various SAUs.

HIGHLIGHTS

- * One of the hallmarks of the year was a two week international workshop on Management Information System for Agricultural Research in collaboration with ISNAR and supported by the Asian Development Bank. Delegates from six countries -- China, Pakistan, Bangladesh, Nepal, Sri Lanka and India participated.
- * Many new programmes were started which include Administrative and Financial Management, Information Management, Personnel Management for the Senior Executives and workshops on Economic Analysis and Returns to the Research Investments.
- * An Outreach Programme in Educational Technology, a unique programme, was launched for the in-service teachers for the upgradation of educational programmes, learning-teaching activities etc.
- * A Workshop on Evaluation of Agro-forestry Education organised with USAID support, provided an opportunity to review the current strategy of Agro-forestry education in the country.
- * A Summer Institute on Instructional Technology with special emphasis on enhancing teaching-learning effectiveness was organised to bring out the need for appropriate planning of instructional systems, and optimum use of instructional resources keeping in view the complexity in the teaching-learning process of modern knowledge.
- * A Travelling Workshop on Educational Technology and Research Management was organised for the first time by the Academy for the teachers of the Assam Agricultural University in their main campus Jorhat at the request of the Vice Chancellor.
- * A Collaborative study with the University of Queensland, Brisbane, Australia, supported by the Australian Council of International Agricultural Research, on the impact of their publication "Nutritional Disorders in Sorghum" was completed. The study has given insight into effective use of resources in bringing out publications that are meant for wider use.
- * The Foundation Course on Agricultural Research Project Management (FCARPM) was modified and reorganised to meet the emerging requirements.

- * The Academy has taken up a new series of publication entitled "Occasional Papers". In this series, the papers critically examine the current and arising issues in the areas of Research Management, Educational Technology, Administration and Management as well as newer approaches to research planning and prioritisation. The first Occasional Paper was on "Curriculum Development for Higher Education in Agriculture: Arising Issues". Likewise, proceedings of various workshops were also published. The revised manual on Field Experience Training was published as "Manual on Field Experience Training for ARS Trainees".

- * The Faculty building, with many of its new facilities, was commissioned to use. A Computer Laboratory comprising a mini frame computer WIPRO S - 386 with 12 terminals in addition to the DCM TURBO 2000 PC-AT, was installed. Three dot matrix and two letter-quality printers besides one plotter were added to this facility. A Self-Aided Instructional Laboratory (SAIL) which provides the trainees to undergo structured and programmed instruction under self-learning situations was another facility added to the Academy during the year. A Video Instructional and Production Laboratory comprising two studios and a control room has started functioning. The studios are equipped with 40XC M3AP cameras, portable table model video cassette recorders, edit controllers, special effects generator, chroma keyer, 8-channel audio mixer, micro computer for generating characters and multiplexer for recording from films and slides.

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Experience

TRAINING PROGRAMMES

The Academy organises several types of training programmes to fulfil the needs of the clientele groups working at various levels of management in the agricultural research and development organisations. Three levels of clientele -- junior, middle and senior -- are identified for training purposes.

The list of training programmes and workshops organised during the year is given below:

Sl. No.	Title of the Programme	Dates	No. of Participants
1.	Workshop on Curriculum Design and Development	April 4-5, 1990	19
2.	Training-cum-workshop on Research Planning for National Agricultural Research Projects	April 16-20, 1990	17
3.	Summer Institute on 'Instructional Technology with Special Emphasis on Enhancing Teaching-learning Effectiveness	June 4-23, 1990	28
4.	Course on Agricultural Research Project Management (CARPM)	July 17-29, 1990	30
5.	Course on Technical Information Management, Research Communication & Project Maintenance in Agriculture	July 31 to Aug. 10, 1990	43
6.	Course on Human Resources Management	Aug. 20 to Sept. 1, 1990	22
7.	Workshop on Returns to Investment in Agricultural Research	Aug. 28-30 1990	29

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8.	Programme on Administrative and Financial Management	Sept. 3-7, 1990	24
9.	Course on Research Management in Home Science	Sept. 11-22, 1990	8
10.	International Workshop on Management Information Systems for Agricultural Research	Sept. 17-29, 1990	41
11.	Foundation Course on Agricultural Research Project Management	Oct. 9 to March 7, 1991	50
12.	FET Coordinators Workshop	Nov. 8, 1991	18
13.	National Workshop on Organisational Management in Educational Institutions	Nov. 19-23, 1990	12
14.	National Workshop on Self-Instructional Systems for Higher Education for Agriculture	Dec. 3-7, 1990	17
15.	Course on Human Resource Management	Dec. 11-22, 1990	14
16.	Workshop on Evaluation of Forestry Education	Dec. 11-13, 1990	22
17.	Travelling Workshop on Educational Technology	Dec. 26-31, 1990	35
18.	Outreach Programme in Educational Technology	March 12-16, 1991	11

Courses for Junior Scientists

A regular course is offered to the scientists recruited to the Agricultural Research Service (ARS) by the ICAR. The training programme, in operation since 1976, had undergone structural changes in concept, duration and details of operation during this period. The experience of the last 14 years and the feedback obtained

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from the participants have been considered in modifying and redesigning the programme to meet the emerging requirements. In the current approach emphasis is laid on pro-active learning and integrated experiential learning, so as to build competency, analysis, synthesis, application and formulation, providing hands-on experience to enable the learner adopt a holistic approach and systems thinking in identifying and managing projects. The course is organised within the frame work of the following objectives:

- * to acquaint the scientists with the general agricultural scenario in the country with reference to research and development,
- * to equip them with techniques in research project management,
- * to give an insight into the rural society and structure,
- * to develop skills in organisational behaviour for high motivation, team work and conflict resolution,
- * to provide training in agricultural communication and transfer of technology,
- * to orient the ARS scientists of the ICAR and the others of National Agricultural Research, Education and Extension Education Systems in the country and
- * to foster a feeling of fraternity among the members of the ARS serving in the different parts of the country.

In order to fulfil the above objectives, the course has been structured in three phases with theoretical and practical classes.

In Phase I, spent at the Academy, instruction on various aspects of project management and national priorities in agriculture are given emphasis, after a detailed exposure to the National Agricultural Research, Education and Extension Systems in India. During Phase II, the trainees are attached in small groups to the Regional Research Stations of the Agricultural Universities or ICAR institutes where they receive practical training in project identification and formulation, technology development process and mode of transfer of technology, besides developing an empathy for farmers and appreciation for rural life. Phase III, spent at the Academy is the follow-up and refinement of the experiences for conceptual development in all related areas like agricultural communication, audio-visual aids, treatment of research findings and presentation of scientific papers. During the period under report, 70 ARS scientists freshly recruited into the service, completed their training.

The following course contents are covered in the training programmes:

I. General Agricultural Scenario in the country

1. NAARM and its Role
2. Agricultural Research Management Process
3. Agroecological Zones and Cropping Systems in the World
4. Agricultural Production -- Potential and Constraints
5. Livestock Production -- Potential and Constraints
6. Impact of Green Revolution -- Strategies for Future

II. Rural Society and Structure

1. Rural Development : Basic Issues and Suggested Policy Frame
2. Rural Leadership and Rural Development Programmes
3. Socio-economic Surveys and Tools of Data Collection for Socio-economic surveys
4. Analysis of Data of Socio-economic Survey for Formulating and preparing Viable Research Projects

III. Research Projects

1. Creativity and Problem Solving
2. Research Programme Planning
3. The Research Project
4. Project Appraisal
5. Project Selection Exercise
6. Project Management Techniques - CPM & PERT
7. Resource Smoothing
8. Uncertainties in Research

IV. Transfer and Adoption of Technology

1. Technology Transfer and Appropriate Technology
2. Institutional Structures for Transfer of Technology
3. Treatment of Research Findings
4. Effective Presentation Techniques in Scientific Communication
5. Communication through AV aids
6. Communication through Mass Media
7. Development Communication in Agriculture
8. Constraint Analysis in Research and Adoption
9. Diffusion Process in Agriculture

V. History, Structure and Organisation of Agricultural Research

1. History and Growth of ICAR
2. Structure and Organisation of ICAR
3. Agricultural Research Service
4. National Agricultural Research Project
5. Farming Systems Research
6. National Agricultural Research Systems in India
7. Systems Approach for Technology Development
8. Brief Review of Agricultural Research Systems in the World
9. Linkages and Coordination between ICAR and Agricultural Universities - Role of ICAR
10. Linkages of ICAR with Other National Research Organisations
11. Research and Extension Systems in Agricultural Universities
12. All India Coordinated Research Projects of ICAR

VI. Organisational Behaviour

1. Interpersonal Relationships
2. Stereotypes

VII. Team Work and Interpersonal Relationships

1. Individual and the Organisation
2. Groups in Organisation
3. Team Work
4. Group Dynamics -- Exercise
5. Motivation and Behaviour
6. Motivation and Work
7. Exercise on Motivation
8. Transactional Analysis
9. Conflicts in Organisations
10. Positive Thinking

VIII. Training in Skills Development

1. Communication Skills
2. Rapid Reading and Comprehension
3. Basics of Computers and Use of Computers in Agricultural Research and Education
4. Statistical Inferences

5. Designs of Experiments
6. Basic Concepts of Linear Programming
7. Proof Reading
8. Production and Printing Process

IX. Economics of Agricultural Development

1. Agricultural and Economic Development with reference to Five Year Plans
2. Agricultural Labour and Employments
3. Agricultural Finance
4. Agricultural Prices
5. Land Systems and Measures of Reform and its impact on Agricultural Productivity

X. Organisation of Research

1. Internal Research Organisations -- ICAR and International collaboration in Research
2. Research Projects Monitoring and Evaluation in ICAR
3. Contribution of ICAR to Agricultural Education Systems in the country
4. Consequences of Technologies in Agriculture

XI. Information Management

1. Sources of Agricultural Information
2. Citation Indexing
3. Management of Personal Documents
4. Book Reviewing in the Sciences and Physical parts of a book
5. Poster Presentation
6. Writing of Scientific Paper
7. Technical Report Writing
8. Writing of Articles for Green and Glory
9. Review of Literature
10. Transfer of Scientific Information
11. On-line Information Services and Electronic Databases

XII. Recent trends in Research

1. Crop Sciences
2. Oil Seeds
3. Dryland Agriculture
4. Pest Management

5. Soil Management
6. Fisheries Management
7. Livestock Management
8. Horticultural Crop
9. Poultry Management
10. Farm Machinery and Power
11. Irrigation and Water Management
12. Energy Management in Agriculture
13. Solar and Wind Energy
14. Biomass and Energy
15. Pollution

XIII. Administration and Accounts

1. General Service Conditions - An Overview
2. Stores Management and Purchase Procedures
3. Project Budgeting and Monitoring
4. Management of Research Stations

XIV. Practicals I

1. Audio-visual aids
2. Treatment of research findings
3. Library

XV. Practicals II

1. Green Revolution Game
2. Book Review presentations
3. Field visit for Field Experience Training

XVI. Visits

1. ICRISAT
2. National Institute of Nutrition
3. National Remote Sensing Agency
4. Other ICAR institutes

XVII. Others

1. Group Seminar presentations
2. Feedback on group seminar presentations
3. Socio-cultural field work

Courses for Middle and Senior Clientele

MANAGEMENT DEVELOPMENT PROGRAMMES

Programme on Administrative and Financial Management (Coordinator : V. Raghavendra Rao)

The programme is organised for senior executives of ICAR, including Deputy Director Generals and Assistant Director Generals at headquarters and Directors of ICAR institutes. This programme is designed to improve the managerial effectiveness of senior executives of ICAR in the fields of "Administrative and Financial Management" by giving them exposure to the concepts in the above areas with reference to Management of Agricultural Research.

The course was organised from September 3 - 7, 1990

The programme dealt with Administrative and Financial Management and covered the following topics:

Administrative efficiency and Productivity, Motivating the Personnel related to administration and Financial Management, Computerisation of office procedures, Authority and Accountability, Legal issues, Labour management, Performance appraisal, Communication conflicts, Disciplinary issues, Vigilance issues, Scope of Finance function in research institutes, Project and Budget Formulation in ICAR institutes, Budget and Resources allocation at ICAR; Approaches to Budgeting - A zero base orientation; Project based monitoring at ICAR institutes; Performance Monitoring at ICAR; MIS for Financial Planning and Monitoring; Panel Discussions - Implementing Improvements in Systems i) Administrative Management, ii) Financial Management.

Twenty four participants comprising four Assistant Director Generals, 15 Directors and five Principal Scientists attended the programme.

The participants discussed the issues related to the Administrative and Financial Management and the following recommendations were made for the consideration of the ICAR.

I. ADMINISTRATIVE MANAGEMENT:

1. A committee, consisting of the senior scientists one from each unit/division of the institute, may be constituted to advise the Director with regard to scientific, technical and administrative matters.
2. In each sub-station an Assistant Administrative Officer and an Assistant Finance and Accounts Officer should be appointed to provide support to the Officer-in-Charge.
3. Major procedural constraints with regard to the following should be removed.
 - a) Purchase procedures should be streamlined, liberalised and made known to all with regard to the imports of equipment; maintenance of vessels and vehicles, etc. Publication of a comprehensive manual outlining all the procedures suggested.
 - b) Director (works) cell should be strengthened so that the works taken up by CPWD are effectively monitored and supervised for quality construction.
4. Delegation of powers: The following administrative functions may be delegated to the Senior Scientists.
 - a) Sanction of leave (casual / earned leave), approval of the tour programmes and tour advances to all the personnel working under them.
 - b) Sanction of contingent expenses for the purchase of chemicals and glassware and essential stores for project activity.
 - c) Purchase of stores material upto Rs.5,000/- at a time.

Modernisation

5. Modernisation of office and standardisation of office procedures may be undertaken with regard to :
 - a) The administrative and financial procedures, records, the maintenance of personnel files and service books, inventory of stores material and the retrieval of information by the use of computers. Necessary software should be developed by ICAR and distributed to the institutes for improving the administrative efficiency in the organisations. Facilities should be provided for uninterrupted power supply to the research organisations.

- b) Purchase of standardised items of stationary and stores material for efficient storage and retrieval of items should be insisted upon and standards laid down by ICAR for its institutes.
 - c) Medical reimbursement should be streamlined. For reducing the load of work and for saving time it is proposed that a fixed amount of money proportionate to the basic salary may be paid to the employees, replacing the existing system of medical reimbursement. Alternatively, ICAR may opt for medical insurance like medicare and pay the premium. This will reduce paper work and reduce unnecessary hassles to the Director in this regard.
6. The skills of the administrative staff can be upgraded by conducting refresher courses, sending them for training as soon as they join the organisation in all areas of the administration for a period of 2-3 months.
 7. Institutionalised training may be given to eligible supporting staff who are in service to enable them to get trade certificates to qualify for promotion to technical posts on 20 per cent quota that is in existence.
 8. Training to be given to the administrative staff in the use of PCs, wordprocessing and may be provided with memory based typewriters.
 9. Scientists should be trained in vigilance matters and then should be asked to look after the duties of Vigilance Officers in the institutes. An updated vigilance manual including rules and procedures, inquiries etc. may be prepared by ICAR and sent to each institute.

Selected administrative staff may be given specialised training for dealing with cases related to disciplinary issues.

10. In order to enhance the productivity of the administrative personnel, it is suggested that two advance increments may be given for every five 'Outstanding' annual assessment reports. Alternatively they can be subjected to periodic assessment, in-service tests conducted and promoted to the next grade.
11. Uniform payscales and avenues for promotion may be given to both administrative and technical personnel.
12. Administrative staff, deployed with scientific units, who have to perform two or more roles like secretarial - accounts, secretarial and record keeping secretarial cum purchase etc., may be given special allowance.

Supporting training/services

13. To deal with legal matters pertaining to ICAR institutes, ICAR should identify regional legal consultants to deal with these cases. Some booklets dealing with the legal problems in the institutes should be published, including the updated court cases relevant to ICAR, for enlightening their management personnel.
14. The scientific/administrative officers of the institutes may be exposed to the relevant labour laws that govern the employment of labour in R and D organisations like ICAR.
15. The Directors' cells in the institutes should be able to provide information with regard to scientific/technical/administrative matters and monitor the progress of the ongoing projects. It should have a minimum staff of at least one Technical Officer assisted by a Personal Assistant/Junior Stenographer and a Diary/Despatch Clerk equipped with a PC, Xerox copier and an electronic typewriter.

II FINANCIAL MANAGEMENT:

Budget

1. A separate budget head, Human Resources Development may be created. The allocation under this pertaining head may be used for Training, participation in workshops and seminars etc.
2. A separate budget head "Committed Expenditure" may be introduced (with appropriate sub-head) to include all expenditure towards annual maintenance and repairs of laboratory building and staff quarters, municipal taxes, electricity, telephone charges and maintenance of equipment.
3. A separate budget head for replacing out-dated equipment/instruments with new ones may be given.
4. Budget heads may be coded for computerisation as per the format developed at ICAR headquarters and the budget and financial resource allocations may be computerised.

5. A separate budgetary provision to be made for the coordinating cells of the project coordinators and powers are to be invested on the project coordinators for the utilisation and reallocation of funds within the programme as per needs.
6. A separate budget given for regional centres/stations of the institute and powers to be vested in the Scientist-in-charge of the station for its utilisation.
7. Project oriented budgeting and the division-wise budgeting to be resorted to in ICAR institutes. Project wise budgeting and allotment of funds should be done before the commencement of the financial year.
8. Director may be empowered to reappropriate plan funds as per the needs of the institute without referring to the Council.

Modernisation and Human Resources Development

9. Incentives for Accounts personnel may be provided to improve their productivity by way of providing better working conditions, sending them for training and giving incentives by way of merit increments.
10. For upgrading the skills of Accounts personnel in-service tests may be conducted for promotional posts; periodic training may be given in relevant rules and regulations at the institute or in recognised centres of training.
11. Modernisation of accounting system may be resorted to by way of computerisation of pay bills, store inventories, classified lists of suppliers and money advances of various kinds.
12. A uniform computer package of programme, encompassing all the requirements of Accounting system may be developed by ICAR for the use of the institute under its control.

Course on Agricultural Research Project Management (CARPM)
(Coordinator : K. Venkateswarlu)

The course is designed to provide exposure to the research project managers in the ICAR institutes to the concepts of research projects and application of 'management approach' while dealing with the problems faced by the agricultural research units. It is offered to the senior research managers heading divisions in the ICAR institutes in the cadre of Principal Scientist and Professors and Heads of Departments in the SAUs. The duration of the course is two weeks. This course was offered once during the period from July 17 - 29, 1990.

Thirty participants attended the course.

The topics covered in the course include:

Project Planning - Importance of Planning Long-term and short-term projects in the management process and systematic project planning.

Project selection - Methods and Criteria used for project selection.

Monitoring and Evaluation - Network methods for time scheduling, control and monitoring and resource smoothing of research projects.

Organisation Structure - Principles of organisation and the importance of technology and environment in organisation design.

Group Dynamics - Characteristics and classification of groups, group decision-making and inter-group problems in organisations.

Leadership - Role of leadership in management and leadership styles suitable to Indian conditions.

Conflicts - Types of conflicts that occur in organisation and the methods used to resolve the conflicts.

Organisational Communication - Meaning, concept and importance of organisational structure, communication behaviour, organisation structure's effect on communication, avoiding resentment, principles of effective listening and barriers to listening.

Mass communication - Television in rural communication, steps in developing a TV programme and rural broadcasting.

Course on Human Resources Management (CHRM)
(Coordinator: M.M. Anwer)

The CHRM is designed to give exposure to the Research Managers to the concepts of human behaviour in organisation and topics related to human resources management in agricultural research. This course is organised for senior scientists heading divisions in the ICAR institutes in the cadres of principal scientist and professors and heads of departments in the SAUs. The duration of the course is two weeks.

The course was organised twice during the period, from August 20 to September 1, 1990 and December 11-22, 1990.

Thirty six participants attended the course which covered the following topics:

Individual and the organisation; Motivation of the Scientists; Interpersonal relationships; Transactional Analysis; Group Dynamics; Leadership; Teamwork; Conflicts; Personnel Policies; Structure and design of Organisation; Organisational Communication; Mass Communication; Research Project Management.

Course on Research Management in Home Science
(Coordinator: A. Gopalam)

The main objective of the course is to expose the Professors and Associate Professors in Home Science Colleges, Agricultural Universities and ICAR institutes to the basic concepts of developing research projects, application of pragmatic and inter-disciplinary management approach to research problems.

The course was conducted from September 11-22, 1990. Eight senior faculty members participated in the course.

The topics covered by the course include:

Research Management Process, Project Planning, Project Evaluation, Basic Elements of Scientific method, Planning for Research Study, Research Organisational Behaviour, Organisation Structure, Group Dynamics, Leadership, Conflicts, Problem-solving Decision-making, Organisational Communication and Mass Communication.

Outreach Programme in Educational Technology (Coordinator : R. Sudarsanam)

The organisational structure in agricultural universities to transmit the knowledge and skills acquired by them by attending workshops and training programmes to their colleagues is weak. This points to the necessity to support the establishment of Educational Technology Cells in every agricultural university for organising the necessary in-service training facilities. For the Cells to function effectively, there is a need to produce the required manpower trained in all aspects of educational technology as applied to higher agricultural education system. It is to serve this need for educationally trained technical manpower that the NAARM is organising an "Outreach Programme in Educational Technology" under the flexible system of distance education. The first Contact Session was organised from March 12 - 16, 1991. Eleven faculty members from SAUs who are engaged in teaching programmes participated.

Objectives

To offer an educational programme covering all aspects of higher agricultural education to members of the faculty in SAUs and ICAR deemed universities to produce appropriately trained persons. These, in turn, could serve the needs of these institutions for (a) organising in-service training programmes for the teachers; (b) undertaking educational research and (c) contributing to the upgradation of educational programmes, learning-teaching activities etc.

Design of the Programme

1. The programme is of one year's duration with 72 modules and two projects. Each module takes up specific classroom based issue under relevant curricular area of study; engages the reader through interactive processes of analysis and synthesis in terms of practice-based principles (praxis) and attempts to impart improved teaching competencies among the trainee-members. This is delivered in eight learning packages, with each package carrying nine modules.
2. The reading materials were prepared specially for the programme, taking into account the classroom experience of the teachers and preparing the outreach lessons in the form of 'Self learning modules' for upgradation of knowledge and skills in all aspects of instruction.
3. These materials are mailed to the trainees for their study, and try-out. The trainees will have to return the response-sheets (feedback) after completing their 'teaching-based' responses.

4. There will be three Contact Sessions - two in the Phase I (one week of orientation in the beginning and another of two weeks duration in the end) and one in the Phase II (two weeks) at the end. These Contact Sessions are compulsory, and are meant to provide training instructional skills, preparation of resources materials like audio tutorials, computer assisted instructional lessons, self-instructional modules, evaluation materials etc. Practicals and Hands on Experiences consist of three contact sessions.

Curricular Themes and Issues:

The programme covers -- Higher Agricultural Education: Philosophy, Administration and Development; Understanding the Learner: Sociological and Systems Theory of Learning; Curriculum (Macro and Micro levels): Modification and Development; Techniques, Methods and Models of Learning-Teaching; Production of Instructional Resources for Classroom use and individualization; Group Methods for Team Work, Research (Field-lab-based) Skills and Creativity, Management of Slow/Gifted learners and Co-curricular activities; Testing, Evaluation and Assessment; HRD and Research in Educational Technology; Classroom-based and Institution-based Projects.

WORKSHOPS

1. Workshop on Curriculum Design and Development (Coordinator: Devesh Kishore)

The workshop was organised from April 4 - 5, 1990 to impart exposure on the principles of Educational Technology in the course and curriculum development to Directors of the Summer Institutes. The workshop was designed mainly to bring about the qualitative improvement in updating teachers, researchers and extension specialists in the latest knowledge and techniques in the respective fields of their specialisation. The course content of the workshop were specifically meant to provide a common forum of interaction with co-professionals, exchange experiences, maintain feedback and to enhance overall relevance of education. One of the objectives of the workshop was also to design a new course or a set of courses by redesigning existing courses in the specific areas covered by Summer Institutes.

Twenty Directors of the Summer Institute participated in the workshop. The participants were explained the expectations regarding the development of the course from the Directors of Summer Institutes.

2. Workshop-cum-Training on Research Planning for NARP (Coordinators : T. Balaguru & S. N. Saha)

A National Workshop-cum-Training programme on research planning for NARP was organised from April 16-20, 1990 for the Directors and Associate Directors of Research from the SAUs. The workshop, attended by 17 participants from 13 agricultural universities, was organised around two major theses -- i) Planning relevant research for specific agro-climatic zones by using the status report and ii) Appropriate research management tools to plan such research activities.

The competence and experience of the speakers as well as the participants greatly contributed to high quality discussions. From the deliberations in the workshop, many useful recommendations related to i) status report preparation ii) relevant research planning and iii) various implementation issues - conceptual, administrative, financial, human resources and policies, have emerged.

Recommendations

i) Status Report Preparation:

- a. Status report should be prepared in time.
- b. During data collection formal and informal methods should be combined
- c. Checking and rechecking of collected data to tally with that of bureau of economic statistics
- d. Adequate attention to identify farming situations during the zone
- e. All scientists should have ready access to status reports

ii) Relevant Research Planning

- a. Information in status report should form the basis for planning relevant research
- b. Concerted efforts in prioritising the research problems using modern methods and allocation of resources
- c. Opening of new sub-stations within the zone should be preceded by consideration of its aspects carefully.
- d. Modern management tools like dendogramming and brainstorming with holistic farming systems perspective should be employed involving all scientists in the station.
- e. In addition to crops inclusion of livestock, agro-meteorology, and agro-forestry should also be considered
- f. Development of time-bound research projects should be encouraged.
- g. Developing research projects in basic science areas utilising assistance from ICAR and World Bank.

iii) Implementation Issues

A. Conceptual Issues

- a. Philosophy and concept of NARP should be clear to all those who are concerned with its actual implementation
- b. Conceiving NARP as a separate elite project is a misunderstanding
- c. All research activities in the regional station need to be integrated
- d. NAARM should continue to organise special programmes for the Directors and Associate Directors of Research to appraise them of the NARP concept.

B. Administrative Issues:

- a. All communication from ICAR should reach Directors of research
- b. Delegating enough authorities to the Associate Directors of Research - commensurating with their responsibilities
- c. Monitoring system needs to be strengthened with the development of suitable proforma for monitoring project activities
- d. Provision of extension of time for the sub project should be considered favourably

- e. Timely actions for removal of administrative bottlenecks and full utilisation if funds allotted.

C. Human Resources Development Issues:

- a. Paying immediate attention for recruiting adequate staff
- b. Development of adequate manpower in areas like agricultural economics, agricultural engineering etc. will help in developing more relevant research projects
- c. Inculcating the spirit of multidisciplinary cooperation to solve agricultural production problems
- d. NAARM should organise special programmes like Training of Trainers for covering more number of scientists within a reasonable time period

D. Policies Issues

- a. The highest authority of the University should be appraised of the difficulties, achievements and the shortcomings during the implementation of NARP
- b. In the zonal meetings state bureaucrats should be involved
- c. Frequent transfer of Associate Directors of Research is detrimental to the project purpose
- d. For planning civil works for the Regional Research Station, all involvement of Associate Directors is necessary
- e. Good libraries with necessary books and scientific periodicals
- f. Organising Management Information System covering personnel projects, funds and other physical facilities should be developed
- g. After ICAR support is withdrawn, continued support of state governments to the zonal research station to work on NARP concept and philosophy is needed.

3. N. C. E. R. T. Workshop

(Coordinators : Jagdeesh C. Kalla & A.K. Sacheti)

The Department of Vocationalisation of Education, NCERT, New Delhi organised a workshop to develop instructional material i.e. two books - Fruit culture, Plant propagation for the use of students offering horticulture vocational course at +2 stage of Education at NAARM, Hyderabad from June 12 - 17, 1990. Twelve eminent horticulturalists of the country from SAUs and ICAR institutes participated in the workshop. The workshop was inaugurated by Dr. K.V. Raman, Director, NAARM. Dr. Jagdeesh C. Kalla, Joint Director of the Academy acted as the Course Director and Dr. A.K. Sacheti of the NCERT, New Delhi acted as Programme Coordinator of the workshop.

Another workshop to review and check the authenticity of the contents of the two books developed by the above workshop was also organised from June 20 - 22, 1990. Books were reviewed by eminent horticulturalists.

International Workshop on Management Information Systems for Agricultural Research

(Coordinators: K.V. Raman & T. Balaguru)

An International Workshop on Management Information Systems was jointly organised by NAARM and ISNAR with the support of Asian Development Bank from September 17 - 29, 1990 for the research managers from the South Asian countries who either use information for decision-making or provide information for such decision makers. Primary objective was to develop and evaluate new approaches to organising and using good information for agricultural research management. The workshop essentially focussed on specific MIS development problems and practical solutions.

Forty one research managers from six South Asian countries -- Bangladesh, People's Republic of China, Nepal, India, Pakistan and Sri Lanka including participants from NAARM, ISNAR and ADB participated in the workshop. The participants had 'hands-on' experience by working on computers with the 'REFLEX' software. The guidelines prepared by ISNAR on PBS/MIS were discussed by different panels comprising scientists from different countries.

After familiarising with basic concepts of PBS/MIS, database management and human resources management, the participants in groups of three undertook a project work to develop PBS/MIS by using a case study. The hands-on experience gave confidence to the participants to develop and use MIS for agricultural research. Participants from respective countries finalised the MIS status paper prepared by ISNAR. As a follow up of this workshop, country wise action plans were prepared to implement PBS/MIS. NAARM has also planned to organise national workshop on MIS for the benefit of scientists working in ICAR institutes and agricultural universities.

The outcome of the deliberations in the workshop include:

1. The concept of PBS/MIS became clear to the participants, and their usefulness for managing the research resources more efficiently and effectively was well appreciated.
2. The hands-on experience with the computer and with the software 'REFLEX' helped the participants to build confidence in developing and using an organised MIS.
3. The panel discussions helped ISNAR to improve the guidelines on PBS/MIS prepared by them based on the participants' suggestions.

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Dr. K.V. Raman, Director, NAARM, addressing the participants of NAARM-ISNAR Workshop. Also seen from left are: Dr. Byron Mook of ISNAR and Dr. Muhammed Mannan of Asian Development Bank.

4. The status of the existing MIS in the South Asian countries were clearly brought out in the workshop.
5. The action plans prepared by the participants will be useful for the individual countries to put into practice the concept of PBS/MIS in their specific country situations after effecting suitable modifications.

Participants at the International Workshop on Management Information Systems for Agricultural Research

BANGLADESH

Kamal Rahim, Chief Scientific Officer and Head, Agricultural Statistics, Bangladesh Rice Research Institute, Ghazipur -- 1701, Bangladesh.

Kazi Nazmul Haque, Senior Scientific Officer, Bangladesh Agricultural Research Institute, Joydebpur, Gazipur, Bangladesh.

Mahbub Ahmed, Senior System Analyst, Bangladesh Agricultural Research Council, Farmgate, Dhaka, 1215, Bangladesh.

MD.Abu Taher, Senior Scientific Officer, Bangladesh Jute Research Institute, Manik Mia Avenue, Dhaka - 1207, Bangladesh.

Nasir Uddin Ahmed, Junior Programmer, Bangladesh Agricultural Research Council, Farmgate, Dhaka, 1215, Bangladesh.

CHINA

Zhao Jiansheng, Project Officer, National Centre for Rural Technology Development, P.O. Box 199-2, Beijing 100036, China.

Liang Qu, Chief, Division of Scientific Research Management, Institute of Application of Atomic Energy, Chinese Academy of Agricultural Sciences, P.O. Box 5109, Beijing 100094, China.

Zhao Xiaohu, Program Official, Animal Science Specialist, National Centre for Rural Technology Development, P.O. Box 199-2, Beijing, 100081, China.

Geng Yu, Project Officer, Scientific Management Department, Chinese Academy of Agricultural Sciences, No. 30 Bai Shi Qiao Road, Beijing 100081, China.

Ye Zhihua, Head, Department of Research Management, Institute of Plant Protection, Chinese Academy of Agricultural Sciences, Beijing 100094, China.

Wang Zhongqiu, Project Officer, Department of Scientific Management, Chinese Academy of Agricultural Sciences, No. 30 Bai Shi Qiao Road, Beijing 100081, China.

NEPAL

Bharat Prasad Upadhyay, Research Program Officer, National Agricultural Research Centre, P.O. Box 1336, Khumaltar, Nepal.

Gokarna Acharya, Manager, Computer Unit, National Agricultural Research Centre, P.O. Box 1336, Khumaltar, Nepal.

Harihar Sharma Ghimire, Personnel Administrative Officer, National Agricultural Research Centre, P.O. Box 1336, Khumaltar, Nepal.

INDIA

P.C. Bose, Information Systems Officer, Agricultural Research Information Centre, Indian Council of Agricultural Research, Dr. K.S. Krishnan Road, Pusa, New Delhi -- 110012, India.

R.B. Gaddagimath, Librarian and Head, Information and Documentation, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad - 500 030, India.

P. Manikandan, Associate Professor, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad - 500 030, India.

S. Krishna Moorthy, Professor, Department of Agricultural Economics, Tamil Nadu Agricultural University, Coimbatore - 641003, Tamil Nadu, India.

A.V. Suriya Rao, Scientist (SG), Central Rice Research Institute, Bidyadhapur, Cuttack - 753 006, India.

K. Vidyasagar Rao, Associate Professor, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad - 500 030, India.

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PAKISTAN

Zubair Alam, Assistant Director (Training), Directorate of NILTA, Pakistan Agricultural Research Council, G- 5/1, P.O. Box 1031, Islamabad, Pakistan.

Abdul Jabbar, Senior Scientific Officer (Projects), Planning Directorate of NILTA, Pakistan Agricultural Research Council, P.O. Box 1031, Islamabad, Pakistan.

Muhammad Sharif, Principal Investigator/Senior Scientific Officer, Agricultural Economic Research Unit (PARC), Ayub Agricultural Research Institute, Jhang Road, Faisalabad, Pakistan.

Sultan Zafar, Scientific Officer (Evaluation), NARC TRG Institute, Islamabad, P.O.NARC, Islamabad, Pakistan.

Nasir F. Usmani, Systems Analyst, In-charge of Computer Unit, Pakistan Agricultural Research Council, P.O. Box 1031, Islamabad, Pakistan.

SRI LANKA

Lalitha Damayanthi Galanihe, Research Officer, Planning Cell, Department of Agriculture, P.O. Box 05, Peradeniya, Sri Lanka.

Telge Sarath Gamini Peiris, Biometrician, Biometry Division, Coconut Research Institute, Lunuwila, Sri Lanka.

Sivali S.E. Ranawana, Research Officer, Veterinary Research Institute, P.O. Box 28 Peradeniya, Sri Lanka.

Segu Mohideen Mohamed Zuhair, Senior Lecturer, Department of Agricultural Economics, University of Peradeniya, Peradeniya, Sri Lanka.

RESOURCE PERSONS

T. Balaguru, Associate Professor, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad - 500 030, India.

Asoka Kumari Basnayake, Research Officer, MIS Coordinator, Council for Agricultural Research Policy, 4th Floor, State Mortgage Bank Building, 269 Galle Road, Colombo, Sri Lanka.

Govert Gijsbers, Research Associate, International Service for National Agricultural Research, 2593 BM The Hague, The Netherlands.

J.A. Thumsiman P. Gunawardene, Agricultural Economist, Planning Cell, Department of Agriculture, P.O. Box 5, Peradeniya, Sri Lanka.

Muhammad Mannan, Senior Project Specialist, Asian Development Bank, Manila, Philippines.

Byron Mook, Senior Research Officer, International Service for National Agricultural Research 2593 BM The Hague, The Netherlands.

Barry L. Nestel, Senior Research Officer, International Service for National Agricultural Research, 2593 BM The Hague, The Netherlands.

K.V. Raman, Director, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad - 500 030, India.

Walter G. Rockwood, Management Publications Consultant, RD 1, Box 53 A, Chelsea, Vermont 05038, USA.

Yolanda M. Tanyag, Science Research Specialist, TDRCD/ODED-RD Philippine Council for Agriculture, Forestry and Natural Resources Research and Development (PCARRD), Los Banos, Laguna, Philippines 3772.

Jagdeesh C. Kalla, Joint Director, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad - 500 030, India.

K.V. Subba Rao, Professor (Retd.), National Academy of Agricultural Research Management, Rajendranagar, Hyderabad - 500 030, India.

Workshop on Returns to Investment in Agricultural Research (Coordinators: Jagdeesh C. Kalla & K.P.C. Rao)

This workshop was organised to delineate methods of measurement of returns to research investments in agriculture, limitations of each method, relationship between prioritisation and rate of return methods; and identification of constraints in the application of the present methodologies.

The workshop discussed the topics which include Agricultural Research System in India - organisational and structural characteristics, costs and benefits, methods of research impact measurement like duality based, project evaluation, ex-ante and

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Dr. Jagdeesh C. Kalla, Jt. Director, and Co-ordinator of the Workshop, addressing the participants of the Workshop on Returns to Investments in Agricultural Research. To his right are Prof. Robert E. Evenson, Facilitator of the Workshop and Dr.K.V.Raman, Director.

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ex-post research productivity measurement methods and limitation of methods, discussion of case studies of Indian Agricultural Research System.

Twenty seven participants from agricultural universities, ICAR institutes and some other organisations attended the workshop. Among the various recommendations of the workshop, significant focus was on the role of NAARM for initiating work on relationship between prioritisation on rate of return to investment to agricultural commodities research.

Workshop on Organisational Management in Educational Institutions

(Coordinator : R. Sudarsanam, Facilitator : Bill Coffindaffer)

The objectives of the workshop were to recognise the process relating to institutional organisation and development, study the principles of management and apply them to development activities in teaching, research and extension education, identify and internalise and leadership strategies of promoting and managing 'change'. Fourteen senior level administrators, Deans of faculties, Directors of Instruction and Research attended the workshop.

The workshop created awareness among the participant 'Managers' on the value and need for professional training in the Art and Science of Management in Educational Institutions.

The recommendations include:

- a) Organisation of additional workshops on "Organisational Management in Educational Institutions"
- b) Development of an "Organisational Management" Course curriculum with specific focus on Indian SAUs and ICAR institutions.
- c) Development of specialised faculty internally for conducting this workshop
- d) A three to five year plan for developing Management Case Studies and Training capsules

A programme on Management of organisational activities to "New Managers" i.e. within the first year of appointment to a management position (at all levels of the organisation)

**Workshop on Self-Instructional System for
Higher Education for Agriculture**
(Coordinator : A. Gopalam, Facilitator: Henry D. Foth)

Objectives:

- * to understand the importance of self-instructional and self-evaluation system as a means for overcoming learning difficulties and also for making Learning-Teaching more efficient,
- * to identify difficult areas of learning and develop suitable models of programmed instruction, CAL and programmed review,
- * to identify the methods of integrating self-learning materials with current learning-teaching practices and
- * to create an awareness of the interactive learning systems in the arising field of agriculture and the use of learning resource centres.

Deans and Professors from Agricultural Universities participated. The disciplines they represented included bio-energy, bio-technology, botany, dairy science, entomology, extension, fisheries, plant pathology, plant physiology, poultry science, textiles and clothing, video production and vegetable crops.

The format for this workshop was built around the presentation and self-instruction followed by workshop sessions during which participants developed self-instruction materials. The excellent facilities developed at the Academy for Self-Instructional programmes were extensively used and the model lessons presented inspired and stimulated participants to develop such materials themselves. It was recommended that teachers from particular disciplines should be brought together in a workshop environment to develop self-instructional units for a given course and these materials be shared.

Workshop on Forestry Faculty Training Sub-Project Evaluation
(Coordinator : S.N. Saha)

The workshop was conducted from December 11 - 13, 1990, to review the Forestry Faculty Training abroad; the effectiveness of the sub-project; to review the forestry education in agricultural universities; job opportunities and employment potential for forestry graduates.

Twenty two participants attended the workshop. About 75 faculty members from different agricultural universities have undergone training in USA . They are engaged in teaching and research in agricultural universities. In some universities, degree programmes are being conducted successfully. The participants reviewed the situation and suggested that the post graduate and Ph.D. programmes in forestry can be started wherever facilities are available. After critical review of several aspects of the sub-project it was appreciated that the forestry training is very much essential in the country.

Seven papers critically discussed the different issues in relevant fields. The workshop was conducted by ICAR under the stewardship of Dr. Nagarcenkar.

Travelling Workshop on Educational Technology and Research Management

(Coordinators : R. Sudarsanam and T. Balaguru)

The Academy for the first time organised a four-day travelling workshop on Educational Technology and Research Management from December 26 - 31, 1990 at Assam Agricultural University, Jorhat, at the request of the Vice-Chancellor, for the teachers of the University in their main campus. The workshop was organised to create an awareness and provide scope for hands-on experience in the use of effective and efficient instructional methods and to develop higher level perspectives for undertaking findings for technical development. The workshop utilised the scientists in the university, who had earlier undergone training at NAARM, Hyderabad, as resource persons. Thirty five participants belonging to various disciplines from the four campuses attended the workshop.

After an initial exposure to various topics related to educational technology and research management by the NAARM faculty, the participants worked on various exercises and developed several models on the topics covered. This workshop helped to build core personnel trained in educational technology and research management in the university. These personnel in turn, will organise similar programmes to other teachers in the university, thus producing a multiplier effect. The workshop was well received by the participants and its usefulness appreciated by the senior executives in the university including the Vice-Chancellor.

Evaluation

Workshop on Field Experience Training (Coordinators : Jagdeesh C. Kalla and VE. Sabarathnam)

A Field Experience Training (FET) workshop for coordinators was organised on November 8, 1990 at the Academy.

The workshop focussed on a number of issues relating to the conduct of FET programmes such as the number of trainees to be deputed to each centre and the procedure for evaluation of the trainees work. Several proposals were made at the workshop. These include: two scientists of the same batch should not undertake survey of the same village to avoid repetition of village selection; token honorarium be given to coordinators.

Eighteen FET coordinators from different parts of the country participated in the workshop.

Summer Institute (Coordinator : Devesh Kishore)

A Summer Institute on Instructional Technology with Special emphasis on Enhancing Teaching-Learning Effectiveness was organised at the instance of ICAR. It was designed to bring out the need for appropriate planning of instructional systems, strategies and optimum use of instructional resources. It helped the participants to understand the process of developing lesson plans, experiential processing models and sharpen their skills in production of effective instructional material such as projection aids, audio & video lessons.

In all, 28 participants were selected including two Deans, five Professors and 10 Associate Professors. Discipline-wise there were 12 from Extension Education, five from Veterinary Colleges, six from Agricultural Colleges, four from Home Science Colleges and one from the Academy.

The teaching programme at the Summer Institute was a blend of lectures, group discussion, role plays, audio visual presentation and case studies.

The programme included topics like instructional technology, role, trends, approaches and techniques, role of instructional aids and teaching, planning and selecting instructional materials, making teaching programmes learning centered, programmed instruction and structured learning, experiential learning process. More than 50 per cent time of the programme was allotted for practicals on preparation of lesson plans, working out experiential processing model, development of instructional materials and preparation of audio-visual lessons. Eminent experts and educationists were invited to handle the sessions in the area of motivation, management of teaching-learning process, effective communication

and psychological learning. Some of the instructional material like video lessons, audio lessons, experiential processing model, etc. developed by participants were displayed.

The analysis indicated that all the participants liked the design and were satisfied with the organisation of the Summer Institute. Most of them mentioned that the practical classes helped them in developing new skills and sharpening their existing skills in use of the instructional materials. They also appreciated the films and video shows on the relevant topics.

Consultancy Services

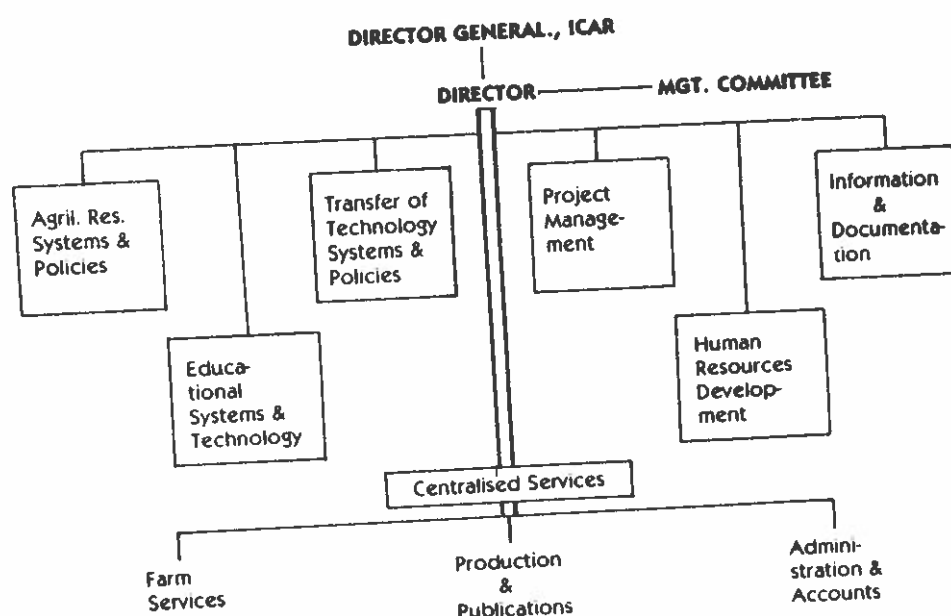
The Academy provides Consultancy Services in the area of Agricultural Research Management and Educational Technology. The Academy guides the SAUs and ICAR institutes in conducting training programmes in these areas. The faculty members of the Academy are specialised in various sub-disciplines and deliver special lectures in institutions which organise training programmes to its own staff. These institutes find the participation of NAARM faculty immensely useful.

Organisational Structure and Salient Activities

The Academy is organised into six subject-matter oriented functional units with a view to impart effective training and conducting research studies in a meaningful way. These are :

1. Agricultural Research Systems and Policies
2. Educational Systems and Technology
3. Transfer of Technology Systems and Policies
4. Project Management
5. Human Resources Development
6. Information and Documentation

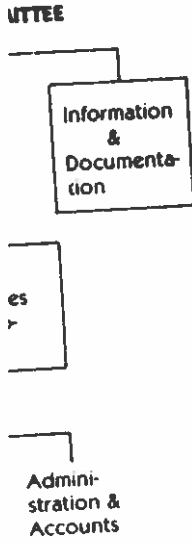
All the above mentioned units function with a strong support from the Centralised Services comprising Farm Services and Agro-Technology Blocks; Production and Publications; Administration and Accounts.



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AGRICULTURAL RESEARCH SYSTEMS AND POLICIES UNIT

Development of Agricultural Sector is conditioned by resilience and capacities of agricultural research system. For, the functions relating to technology generation, gathering of information and procedures involved in agrarian research system would be used jointly for agricultural education, training, extension and client support to bring about desired changes in development strategies. The Agricultural Research Systems and Policies Unit studies the different systems of agricultural research in order to strengthen the various research systems that are essential for agricultural development.

Objectives

- * to study issues on prioritisation of research in Agricultural Research Systems and to assist in development of appropriate science policies,
- * to build agricultural resource and research databases for developing suitable research plans and efficient management of National Agricultural Research Systems,
- * to develop systems approach integrating indigenous knowledge with relevant technologies for varied farming situations and constraints,
- * to study national as well as International Agricultural Research Systems and suggest suitable policy alternatives for improvement,
- * to undertake studies on the socio-economic impact of resource allocation and investment in agricultural research and
- * to undertake studies on futurology in agriculture and technological forecasting and assessment.

Activities

Research Projects

1. "Development of Database on Crops, Forestry Animal and Fishery Resources in National Agricultural Research System".
(P. Kumar)

This project was initiated to analyse and document long-term trends in the constituents of database on crops, livestock, forestry and fishery; to provide pre-cooked basic information on construction of database for policy formulations at national and regional levels; to identify and delineate elements of future strategies for optimisation of agricultural production.

Data has been collected on various parameters pertaining to livestock and fisheries sectors.

2. "Developing suitable Management Information Systems (MIS) for the National Agricultural Research Systems"
(T. Balaguru and K. Vidyasagar Rao)

This project was initiated to develop an organised MIS as a management tool for efficient utilisation of the resources in the research organisations. A pilot study was located in Hyderabad. Data on research projects, personnel and finance were collected through standard proformae and personnel discussion. The data collected are being analysed and the MIS will be developed using the software 'REFLEX'. This case study will be utilised in the forthcoming national workshop on MIS at NAARM.

After further expansion, this project is now being handled by a committee consisting of Jagdeesh C. Kalla, T. Balaguru, P. Manikandan and K. Vidyasagar Rao.

3. "Evolving a Suitable Methodological Framework for priority setting in Agricultural Research".
(Jagdeesh C. Kalla and T. Balaguru)

To evolve a methodological framework for prioritising research, data on projects undertaken by the scientists working at Assam Agricultural University are being collected. On the basis of the collected data, a case study will be prepared to illustrate the application of various priority setting methods such as congruence, DRC Ratios, Checklist, Scoring and Benefit/Cost Analysis.

4. "Patterns of Research Resource Allocation in Agricultural Universities".
(K.P.C.Rao and Jagdeesh C. Kalla)

Objectives: To find out the present patterns of allocation of research resources among disciplines / commodity groups in different agricultural universities and their regional research stations.

To suggest changes in resources allocation among disciplines / commodity groups based on some sound allocation criteria.

Work completed : This project envisaged only some initial preparations during the year. As it was to be a networking project, it was intended to initiate correspondence with the Vice-Chancellors.

5. "Assessment of returns to expenditure on research and development of production technologies for rice-fallow pulses and chillies in Andhra Pradesh".

(K.P.C. Rao)

Objectives: To estimate the rate of return on investments made on specific commodity research programmes such as blackgram and greengram in rice-fallows and chillies in Andhra Pradesh.

To measure the individual and joint impacts of research and extension on the productivity of blackgram and greengram in rice-fallows and chillies in Andhra Pradesh.

Work completed: Literature pertaining to the topic was reviewed in detail. A tour was undertaken for a period of 10 days to collect opinions about new technologies in Guntur and Krishna Districts. Data on research costs, extension projects and to elicit farmers research costs were collected for the period 1971-72 to 1989-90. Met with a partial success only with respect to data on extension projects. Secondary data on crop cuts were collected for the year 1989-90 from the Directorate of Economics and Statistics, Government of Andhra Pradesh, Hyderabad. The collected data was analysed and the results are as follows:

Rice - fallows pulses :

The average area under rabi blackgram in Andhra Pradesh doubled between the triennia of 1979-82 and 1986-89. With the productivity also rapidly increasing from 416 Kg in 1979-80 to 999 Kg in 1988-89, the production of rabi blackgram more than quadrupled in less than a decades time. But in case of rabi greengram, there was a decline in the area, production and productivity.

Advancing of sowing date of blackgram in rice-fallows facilitated by shorter duration rice varieties in the kharif, higher seed rate, timely plant protection, availability of disease-resistant new varieties etc., have all contributed to the increases in productivity of rabi blackgram. In order to find out the change and effect relationship between research expenditures and productivity growth, a regression equation was estimated with the productivity of rabi blackgram in Andhra Pradesh as the dependent variable and weighted research expenditures on pulses at Lam Farm, Guntur at constant 1970-71 prices with the assumption of 2-2-2-2 lag structure as the explanatory variable. The estimated equation for the period 1979-80 to 1988-89 was :

$$\begin{array}{lll} \text{Productivity} = 428 + 0.7103 \text{ Res. variable} & R^2 = 0.8144 \\ * 't' - \text{value} & (5.924)* & M = 10 \end{array}$$

This estimated relationship was highly significant with a good explanatory power. It suggested that the productivity increased by 0.71 kg for every thousand rupees increase in the weighted research expenditures variable.

To work out a benefit-cost ratio, the research costs incurred between 1971-72 and 1988-89 were converted into 1988-89 rupees using the general price index (1970-71 = 100). Following Evenson and Mc Kinsey's study, the extension costs were assumed to be three times the research costs. The expenditures incurred under National Pulses Development Project were also added up. The total cost of research and development worked out to Rs. 37.16 million. A very conservative estimate of the benefits amounted to Rs.231.2 million, yielding a benefit cost-ratio of 6.22.

Chillies:

It is an important condiment crop in Andhra Pradesh. The acreage and production of the crop fluctuated widely over the years, leading to tremendous swings in the price of the product as well. But there is a discernible increase in the productivity of the crop as result of the new production technologies evolved by the Lam Research Station, Guntur.

A research variable was formed by adopting a 2-2-2-2 lag structure to the research expenditures after deflating them with the general price index (1970-71 = 100). Productivity (average yield at the state level) was regressed upon the research variable with 10 observations from 1979-80 to 1988-89. The estimated equation was :

$$\begin{array}{lll} \text{Yield} = -422.56 + 4.7135 \text{ Res. Exp} & R^2 = 0.8162 \\ * 't' \text{ value} & (5.961)* & m = 10 \end{array}$$

The above equation was statistically significant with a fairly good explanatory power. It implied that the productivity of chillies showed an increase of 4.71 kg for every 1000 - rupee increase in the weighted research expenditures variable.

To estimate a benefit-cost ratio, the research costs incurred at Lam Research Station since 1971-72 were evaluated at 1988-89 prices, using the general Price Index (1970-71 = 100). Since the research scheme was in place since 1928-29, it was assumed that the research costs averaged about Rs.300 thousand per year at 1988-89 prices. The total research expenditures with this liberal assumption aggregated to Rs. 19.73 million. The extension costs were assumed to be three-times the research costs, bringing the total research and development costs upto Rs. 78.92 million.

The yield advantage due to new varieties of chillies was estimated from the data on crop cutting experiments of 1989-90 to be 7.7 quintals per hectare. At an average price of Rs. 650/- per quintal prevalent in 1989-90, the gross benefit due to new varieties was Rs.5,005/- per hectare. According to the data elicited from

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the farmers, the new varieties entailed an additional cost of cultivation of Rs. 3,950/- per hectare, leaving a net benefit of Rs. 1,055/- per hectare due to new varieties. Assuming that the Lam varieties have gone to 50 per cent of the crop area with and an adoption rate of 60 per cent, the annual benefits were estimated to be Rs. 53.96 million. It represented a very high annual rate of 68.37 per cent on the investments made on research and development.

6. "Policy analysis in involvement of Technology-Advocates in development of crop technologies".
(VE. Sabarathnam)

Objectives:

- * to identify the different types of advocates of crop technology who could be involved in development of crop technology,
- * to locate the specific steps in technology development at which different advocates could be involved in the initial process of technology development
- * role analysis of the different advocates in the development of crop technologies
- * to find out the problem faced by the different advocates of crop technology in their involvement in technology development process and
- * to suggest the policy-mix for overcoming problem in involving technology advocates in technology development.

Relevant literature has been collected. The scheduled for collecting data from the technology developers is under preparation.

7. "Development of Methodology for Selection of Psychological Factors as Independent Variables".
(VE. Sabarathnam)

In order to select the most appropriate psychological factors as independent variables for a study, an Influence Quotient was considered necessary because arbitrary selection of some variables by intuition for a study may end up with frustrating results. This influence quotient will help scientists to select the psychological factors in a scientific way for any type of psychological study to get significant research results. The quotient had reliability of 0.685 which was significant at one per cent level.

8. "Estimation of Research Climate and its Relation with Efficiency of Scientists"
(VE. Sabarathnam)

The research climate in a research system plays an important role increasing the efficiency of the scientists. However, no instrument was available to measure the research climate scientifically. Hence, an attempt was made to develop a research climate quotient with 18 indicators. The quotient has reliability of 0.85 and a validity of 0.61. These values were significant at one per cent level. This indicates the reliability and validity of the research climate quotient. This can be used by any research organisation to assess the climate that is existing in a research organisation for conducting research so that corrective measures can be taken up to improve the research climate. The relationship between the research climate and the efficiency in the dryland research system was also estimated and it was found that the climate contributed only meager because of the intervening nature of the variable-research climate.

9. "Estimation of Evaluation of Students' performance at different levels".
(K. Vidyasagar Rao)

During the course of career development in education it is a preconception, that the past knowledge gained have direct or indirect impact on present and future pursuit for learning. The nature and extent of the relationship however, vary from individual to individual depending on one's own IQ. The present study aimed to quantify the inter-relationship between performances at various stages from the observed data. To start with, the study was confined to the graduate level in agricultural and the data on OGPA and year-wise GPA of four batches (81-85 to 84-88) along with the marks obtained at secondary school (SSC) and undergraduate level (Intermediate) were collected from APAU, Rajendranagar, Hyderabad.

10. "Development of Database on Crops, Forestry, Animal and Fishery Resources in National Agricultural Research Systems".
(K. Vidyasagar Rao and P. Kumar)

The project aims at analysing and documenting long-term trends in the constituents of database on crops, livestock, forestry and fisheries. The useful and relevant parameters of various crops and animals were identified and listed. In case of forestry and fisheries, identification of important parameters are in progress with the consultation of the respective area specialists. After finalising the formats the work on collection of data will be undertaken.

Workshops

1. Workshop-cum-Training on Research Planning for National Agricultural Research Project was organised from April 16 - 20, 1990.
2. Workshop on FET Coordinators from State Agricultural Universities was organised on November 8, 1990.
3. Workshop on Returns to investments in Agricultural Research was organised from August 20 - September 1, 1990.
4. NAARM - ISNAR International Workshop on Management Information System for Agricultural Research was organised from September 17 - 29, 1990.

Course Material Developed

1. Economic Evaluation of Agricultural Research Projects
2. Dimensions of Forest Policy in India
3. Agricultural and Economic Development with reference to Five Year Plans
4. Research Prioritisation in Agriculture
5. Sustainability issues in Indian Farming
6. Management of Job Stress
7. Involvement of Rural leaders in implementing developmental plans
8. Socio-economic considerations in optimal land uses in semi-arid regions in 'Technologies for wasteland Development'.

EDUCATIONAL SYSTEMS AND TECHNOLOGY UNIT

The primary function of this Unit is to render assistance to the SAUs and ICAR institutes dealing with education and provide support to develop better teaching techniques. Education technologists interact closely with Subject Matter Specialists to develop viable and useful courses and teaching techniques.

Objectives

- * to organise teaching and training programmes in higher agricultural education with special reference to curriculum development, instructional systems, production of learning resource materials, evaluation and assessment, and educational administration,
- * to undertake research projects and case studies for evolving more effective and efficient approaches, methods and models in higher agricultural education system,
- * to develop information database on education system in SAUs and ICAR institutes,
- * to provide technical support to SAUs and ICAR institutes on educational matters to enhance their teaching-learning effectiveness and
- * to establish linkages with international bodies imparting higher agricultural education.

Activities

Workshops

1. Workshop on Curriculum Design and Development was organised from April 4 - 5, 1990 for the Directors of Summer Institutes from ICAR institutes and Agricultural Universities.
2. Summer Institute on Instructional Technology with special emphasis on enhancing teaching-learning effectiveness was organised from June 4 - 23, 1990 for Deans, Professors and Associate Professors and Senior Scientists of ICAR institutes and Agricultural Universities.
3. Workshop on Organisational Management in Educational Institutions was organised from November 19 - 23, 1990 for Senior level Administrators, Deans of Faculties, Directors of Instruction and Research.

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4. Workshop on Self Instructional Systems for Higher Education for Agriculture was organised from December 3 - 7, 1990 for Deans, Professors and Associate Professors of Agricultural Universities.
5. Travelling Workshop on Educational Technology and Research Management was organised at Assam Agricultural University, Jorhat from December 26 - 31, 1990.

Courses

Course on Research Management in Home Science was organised from September 11 - 22, 1990 for Professors and Associate Professors in Home Science Colleges and Agricultural Universities and ICAR institutes.

Research Projects

1. "Production of learning-centred Resource Materials"
(Devesh Kishore, R. Sudarsanam, K.M. Reddy and A. Gopalam)

A computer aided instructional programme, a video tape tutorial and a tape slide presentation were prepared on Plant Pigments. The same is being pre-tested. The efforts are being made to produce video lessons and structured learning materials in the area of information processing, computer and communication.

2. "Outreach Programme in Educational Technology"
(R.Sudarsanam, Devesh Kishore, K.M. Reddy, A. Gopalam and Jagannadham Challa and R.B.Gaddagimath)

The Project on Outreach Programme in Educational Technology leading to Postgraduate Diploma in Educational Technology for Agriculture was launched on March 12, 1991. The main objective of this project is to offer an educational programme covering all aspects of Higher Agricultural Education to members of Faculty in the SAUs and ICAR deemed universities to produce appropriately trained persons who could serve the needs of these institutions for (a) organising in-service training programmes for the teachers; (b) undertaking educational research and (c) contributing to the upgradation of educational programmes, learning-teaching activities etc.

The first contact session was organised from March 12 - 16, 1991. Eleven faculty members from the SAUs and ICAR institutes who are engaged in teaching programmes participated.

UNDP Advanced Centre on Agricultural Education and Research Management

An Advanced Centre on Agricultural Education and Research Management was established in 1987 with FAO-UNDP-UNESCO support. The centre organises workshops, seminars, training programmes and courses in educational technology and agricultural research project management. The Centre has acquired sophisticated equipment to impart effective training in these areas. Several faculty members have received advanced training in various institutes and universities abroad, and a number of overseas consultants visited the Academy to share their expertise with the faculty, under their support.

Henry D. Foth, Professor Emeritus, Michigan State University East Lansing, Michigan, USA was at the Academy from November 7 to December 15, 1990 and held discussions with the Director and the faculty regarding training programmes offered at the Academy. He also acted as the facilitator of the workshop, "Self-Instructional System for Higher Education for Agriculture", held from December 3-7, 1990.

Dr. Bill Coffindaffer, consultant to the FAO/UNDP Advanced Centre for Educational and Research Management was at the Academy from October 27 to November 28, 1990 and held discussions with the Director and faculty members of the Academy. He was also facilitator of the workshop, "Organisational Management in Educational Institutions", held from November 19-23, 1990.

Under the UNDP Project on Advanced Centre for Postgraduate Agricultural Research and Education, four faculty members completed advance study in various institutions and universities in USA.

TRANSFER OF TECHNOLOGY SYSTEMS AND POLICIES UNIT

Development of appropriate transfer of technology systems is important to improve the productivity of researchers and farmers. The effectiveness of Transfer of Technology systems depends on various factors such as managerial efficiency of the managers, span of control, communication strategies, planning and execution. Further, the informal and institutional linkages are the key determinants of success of the TOT systems. Identifying these linkages is a major mandate of this unit.

Objectives:

- * to study the problems and programmes of technology generation and evolve tools for this assessment in transfer of technology,
- * to identify the managerial constraints in existing systems and develop suitable and integrated approaches for effective transfer of technology,
- * to suggest suitable policy framework for effective implementation of technology and
- * to evolve an efficient agricultural communication system to support the aforesaid activities.

Research Projects

1. "Documentation of various Indigenous knowledge systems in Agriculture" (R. Chittiraichelvan, R.K.Samanta, A.U.Gadewar and Jagdeesh C. Kalla)

Objectives:

- * to review and document various indigenous knowledge systems in the area of agriculture and allied fields,
- * to find out cost-benefit ratios of indigenous technical knowledge systems and
- * to assess the feasibility of incorporating indigenous knowledge in technology generation, transmission and utilisation.

Report: Work is in progress

2. "Evaluation of Use and Impact of Sorghum Publication - Nutritional Disorders of Sorghum"

(K.V. Raman and A.U.Gadewar)

Objectives:

- * to find out the extent of distribution of sorghum publication throughout India with particular focus on Sorghum growing areas,
- * to find out who are recipients of publication and
- * to ascertain its usefulness / impact among the researchers, professionals in teaching and extension personnel.

Report : The questionnaires were sent to Directors of Research and Directors of Extension of SAUs, Incharge, Sorghum Research Stations in India, Heads of the Division of Soil Science, Agronomy of the SAUs, the Librarians of SAUs and institutes of ICAR, and Project Directors and Directors of ICAR institutes where sorghum component is primary or secondary. The standard research methodology was followed in developing the questionnaire.

The publication has been well received by the sorghum scientists at various institutes / agricultural universities in India engaged in sorghum research. It could generate discussion among the sorghum scientists belonging to different disciplines but working for sorghum improvement. The plant analysis, soil analysis, visible deficiency symptoms, toxicity were some of the significant items figured in discussion. Also, doubts were cast on the possibility of using nutritional deficiency visible symptoms for accurate diagnosis for nutrition deficiency.

The scientists and teaching faculty indicated that the publication is of high standard and useful in both research and teaching profession. Additional encouraging dimension of the publication was its capacity to enthuse inter-disciplinary team approach on sorghum research.

Scientists as well as teaching faculty indicated that the overall sorghum production would go up by 10 - 20 per cent in case the information contained in the publication is put to use in field and lab conditions.

Course Materials / Lecture Notes developed

1. Development Communication for Agriculture.
2. Technology Transfer and Appropriate Technology.

- Nutritional

throughout India

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3. Rural Development : Basic Issues and Suggested Policy Frame.
4. Effective Presentation Techniques for Scientific Communication.
5. Management of Rural Development Projects.
6. Role Play Exercises.
7. Institutional Structure for Technology Transfer.
8. Indigenous Technical Knowledge Use in Agriculture.
9. Delphi and AHP for Technology Assessment.
10. Management of Agricultural Development Programmes.
11. Commuication through mass media.
12. Communication technologies for the future.
13. Status Report Development Plans.
14. Interactive multi-media .

Inhouse programme:

To sensitise the NAARM Faculty in using video technology in their regular programmes, an inhouse programme on usage of video was executed from January 15-19, 1991. Experience of handling VTR and video camera was provided to the participating faculty members. Different kinds of shots and terminologies in video use were explained. An exercise, "Video Scavenger Hunt" provided an opportunity to gather images. Visual articulation, message development and creating storyboard were dealt in detail. The participating faculty were divided into two groups and using brainstorming techniques, ideas were put together to produce two short video films -- "Litter-free Campus" and "NAARM Agrotechnology Demonstration Block __ A Show-window". Image gathering for these tapes was done by the faculty members. Editing of the logged tapes was also carried out. The inhouse programme resulted in gaining hands-on experience on production of video materials.

PROJECT MANAGEMENT UNIT

This unit is designed to integrate complex efforts with regard to planning, scheduling, directing and controlling Institute's resources for a project established with specific goals and objectives. It provides an individual research worker the necessary tools and techniques which can help him find proven and workable solutions to problems that may occur in the management of a project.

Objectives

- * to evolve an appropriate methodology for identifying relevant Agricultural Research Projects in a given thrust area and to develop criteria for their appraisal,
- * To develop and formulate a holistic/systems approach for project planning and management,
- * to study issues, develop case studies and to evolve suitable tools for carrying out the activities related to project cycle more effectively and,
- * to develop a database system on Agricultural Research Projects needed to fulfil the above objectives.

Activities

The unit has participated in conduct of training programmes for middle and senior-level scientists and foundation courses for direct recruits in ARS in the area of Project Management. It carried out case studies on management of All India Coordinated Research Projects and National Agricultural Research Projects to improve the mechanism of implementation, monitoring and control of the projects.

Research work is being undertaken in the areas of Project Selection, Management of Research Stations through the use of Project Management Techniques / Networks.

Programme on Administrative and Financial Management

This programme was organised from September 3-7, 1990, for the senior executives including Deputy Director Generals, Assistant Director Generals and Directors of ICAR institutes.

Course on Agricultural Research Project Management

This course was organised from July 17-28, 1990 for senior faculty members and senior scientists of ICAR institutes.

Research Projects

1. "Development of Appropriate Criteria for Project Appraisal in Basic & Applied Research in Agriculture".
(V. Raghavendra Rao)

Objectives:

- * to identify the criteria to be used for project selection, and technical merit of the projects in various disciplines of agriculture, fisheries and animal sciences,
- * to assign weights and the scoring techniques for the identified criteria,
- * to examine and develop the quantitative and qualitative rating techniques for project selection to be used by the evaluation teams and
- * to develop criteria for the selection of pure basic research projects.

Workdone: The criteria identified by various groups of senior scientists and the Directors of ICAR institutes who attended the programmes at NAARM are being collected and compiled for developing an instrument.

2. "Management of Agricultural Research Stations -- RRS, Lam and DRR, Rajendranagar".
(K. Venkateshwarlu)

Objectives:

- * to study the development process of Agricultural Research Station,
- * to study the management of land, labour, capital, crops, irrigation system, fertility management, green house, gardens, workshops research projects and transfer of technology and

- * to study the operational constraints.

Data was collected by using questionnaires and interview. It is being analysed.

3. "Development of cases on Monitoring and Evaluation of Research Projects in ICAR institutes".

(S.N. Saha)

Work under progress.

4. "Performance Evaluation of NARP sub-projects.

(S.N. Saha)

Work under progress.

Materials Developed

1. Conference Management.
2. Role of Technical Cell in monitoring research projects for budgets.
3. Zero base budgeting.
4. Research poster sessions at departmental level.
5. Experiential lecturettes.
6. Facilitation skills.
7. Counselling skills.

HUMAN RESOURCES DEVELOPMENT UNIT

The major thrust of this Unit is in the use of experiential learning. Structured experience for training in human behaviour is extensively used for self improvement by trainees, both in the theory and in the organisational behaviour practicals. The concern in all the training programmes is the understanding of human behaviour with a view to changing direct and control behaviour. Emphasis is on group behaviour in addition to individual behaviour. There is a constant endeavour to bring about a general appreciation on the importance and usefulness of effective management of human resources, the most important resource of any organisation.

Objectives

- * to evolve effective and efficient approaches, methods and models in different areas of human resources development as interpersonal relationships, motivation, group dynamics, team-building, conflict management, leadership, organisational climate, institution-building and personnel management,
- * to develop information database on various aspects of personnel management in the ICAR institutes and SAUs for use in manpower planning, recruitment, training, performance appraisal and career development and
- * to undertake systematic review and study of management problems of ICAR institutes and SAUs.

Activities

Research Projects & Experiments

1. A study on culture of trust in research organisations was conducted with the following objectives.
 - * to determine the extent of interpersonal trust/distrust among research scientists in different ICAR institutes,
 - * to identify the causes of distrust among the scientist and
 - * to arrive at alternative action to develop trust among scientists.

Data was collected from the scientists of ICAR institutes by using questionnaires and is being analysed.

2. A baseline survey on expectations and perceptions of senior scientists on 'psychological contract' between scientists and organisation was initiated for participants of various senior level courses by way of Response Sheets. The information thus obtained is being compiled and analysed for identifying areas of personal conflicts between individuals and the organisation.
3. Two laboratory exercises towards developing interpersonal skills team building and inter-group trust levels among both junior and senior scientists have been experimented during the year. There are:

1. Tower building game for supervisory skill development.

2. Scientists dilemma game for trust building and decision making.

These two simulation games have been developed for and tested on the foundation course participants during this year.

Course/Programmes

Two Courses on Human Resources Management were conducted on August 20 to September 1, 1990 and December 11-22, 1990 for senior scientists of ICAR institutes and Professors of agricultural universities.

Resource Materials developed for training programmes

1. Leadership.
2. Transactional Analysis.
3. Structure and design of Organisation.
4. Recent trends in Fisheries development in India.
5. Socialisation.
6. Reflective observation Interdisciplinary perceptions.
7. Individual and Organisation.
8. Understanding group dynamics.
9. Group work and group dynamics for learning.
10. Motivating students for learning.
11. Guidelines for presenting scientific papers.
12. Special coaching, Enrichment and compensatory techniques in learning for slow and gifted learners -- OPET lesson.
13. Career Counselling: Prospects and Processes -- OPET Lesson.
14. Group discussion techniques.
15. Human Resources Development and Education.

New topics developed and delivered for training programmes

1. Excellence as a way of life.
2. Electives-Speed Reading and Comprehension.
3. Science and Technology as part of culture.
4. Ethics, morals and Intellectual property rights.
5. Management of safety aspects in Agricultural Research.
6. Management of job stress and emotions.
7. Selective observations in Interdisciplinary perceptions.
8. Making practicals learning oriented.
9. Motivating students for learning.
10. Human Resources Development and Education.
11. Pollution.

INFORMATION AND DOCUMENTATION UNIT

Information and Documentation form the most important inputs in any scientific activity. Organised scientific information base is important for decision-making at all levels. Personal scientific information management and retrieval is necessary for every research worker. The Information and Documentation Unit provides faculty support for all the training programmes of the Academy in addition to maintaining an excellent Library and Documentation Centre.

Objectives

- * to build a resource Library and Documentation Centre on agricultural research management and educational technology,
- * to establish an information network of agricultural libraries in India,
- * to set up a clearing house for user education/instruction, research on documentation and scientific information management and
- * to assist in developing a policy framework for modernisation of libraries through research on and feedback from the present systems.

Library

The Unit has established a well-organised specialised library. The resource library holds documents on Agricultural Research Management, Science Policy, Planning, Futurology, Forecasting, Project Management, Educational Technology, Technology Systems and Policies, Women Studies, Research Communication, Information Management, Computer Technology, etc. International Corner and Foreign Tour Reports Section is a unique feature. The Library now holds about 21,000 documents and about 425 journals.

Reprographic Services

Reprographic section of the unit is equipped with two Finesse 4515, one Nashua one UBIX 18002 and Toshiba BD 704 plain paper copiers and three scanning machines for meeting the photocopying and reproduction facilities. About 2,76,874 photocopies were made from the plain paper copiers and 180 faxils were cut from the scanning machines. Micrographic section has one Minolta RP 505 and AARQUE microfilming unit. Microfilming unit is being established and will be used for inhouse requirements and other sister institutions.

Computerisation

Computerisation of the library holding is in progress and half of the entire collection is accessible through computer. The information and Documentation Unit has PCAT (40m) and UNESCO's CDS/ISIS package version 2.32 is used for information storage and retrieval. Three separate bases for CAT (books and serials) ARMA (Agricultural Research Management Abstracts) and FTR (Foreign Tour Report) were developed.

Information Resource Lab

Information technology has made a great impact on conduct of research. Current range of CDROM products have made the distance as no barrier for access to information. It is planned to procure many of the useful agricultural databases on CDROM which would be used in training and providing series. The CDROM work station is being established and plans are on for networking the user services.

Faculty Support

Scientific information management is an integral part of almost all the training programmes organised at the Academy. Personal documentation, Fact finding techniques, Review/ Research/ Technical paper writing, Frontiers of information services, Science Citation Index, Science mapping, Research Front specialities in science, study and reading skills, information management tools use in curriculum development and career planning and other topics related to bibliographic and user education were covered.

Activities

Course on Information Management Research Communication and Project Maintenance in Agriculture was organised from July 31 - August 10, 1990 for the faculty, scientists and technical personnel engaged in handling these activities.

Topics relating to Scientific Communication, Publishing and Refereeing, Project Management, Computer Application in Research, Group Dynamics and Leadership, Anticipatory Research Management and Forecasting, DBMS etc. were covered.

Research Projects

1. "Research Productivity of Indian Agricultural Research and Educational Institutes".

(R.B. Gaddagimath and T. Shanmughanathan)

Objectives:

- * to measure the research productivity of ICAR institutes and SAUs,
- * to measure the research output in the different fields of agriculture,
- * to measure the research output in different commodities,
- * to know the authorship pattern of Indian agricultural scientists,
- * to identify the leading journals in the different fields of agriculture and
- * to ascertain the trend of research over a period of time.

After completing the literature collection and study of the subject, the methodology for data collection and analysis was finalised. To begin with the 30,000 references available from Indian National Agricultural Bibliography were grouped commodity-wise. Against each commodity the total number of publications were tabulated subject-wise. The tabulated data is being fed into the computer using LOTUS 1-2-3 package for analysis. Other variables from the references like the name of the journal, author's affiliation etc. are also being added to derive different types of results.

2. "Significant, impact and core periodicals among Indian Agricultural Scientists".
(R.B. Gaddagimath, Y.V. Siva Prasad and T. Shanmughanathan)

Objectives:

- * to identify the significant journals among Indian Agricultural Scientists,
- * to measure the impact of Indian Agricultural Journals based on SCI, CC and other database and
- * to compile a handbook of instructions to authors based on the significant journals of Indian Agricultural Scientists.

Some of the core journals in different fields of agriculture were identified using SCI, CC and other Database list of journals considered core and citation impact was prepared.

A questionnaire was devised to solicit the preferences of the journals by the agricultural scientists working in ICAR institutes. After analysing both the lists the significant journals would be identified and a handbook of instructions to contributors will be compiled.

3. "Directory of Indian Agricultural Library and Information Professionals".
(R.B. Gaddagimath, P.V. Nirmala and Y. Dhana Lakshmi)

Objectives:

- * to compile a comprehensive directory of agricultural library and information personnel in India and
- * to enable identify the contact persons in different subject areas and establish resource sharing.

The different types of agricultural libraries in India were identified and a database consisting of the addresses of the libraries was created.

A questionnaire to collect data was devised and are being mailed to the libraries.

Materials Developed

1. Notes taking and notes making
2. Information networking for NARP
3. Project ALICE
4. Study skills and learning assessment through posters
5. Citation Analysis and Research fronts for education and research management
6. Poster sessions at scientific meetings

Agricultural Research Management Abstracts (ARMA)

This is a unique half- yearly secondary publication brought out by the Academy broadly covering the areas related to agricultural systems and policies, project management, transfer of technology, educational technology, agricultural economics and information management. About 400 scientific and technical periodicals are scanned and significant papers are indexed and abstracted. The computer database on research management was developed for an effective storage and retrieval for the benefit of research managers, educationalists and decision-makers. About 700 copies were distributed for organisations involved in agricultural research, administration, management and policy making. At present, the publication is distributed on exchange basis. Reprographic facilities for the original documents appearing in the abstracts are also offered at nominal cost.

CENTRALISED SERVICES

PRODUCTION AND PUBLICATION UNIT

The Academy has taken up several publications on its own to help the research managers to keep abreast with the latest techniques of management. The Academy itself publishes the materials developed by the faculty for training purposes.

To facilitate the production and publication activities, the Academy is equipped with the Desk Top Publishing system from Apple Macintosh, Reprographic Unit comprising Offset printing press (Swift 300 Deluxe), Plate-making and Binding systems. The unit also has screen printing facility.

Some of the publications printed at the Academy are mentioned below:

1. Course materials and reading materials for each course conducted by the Academy.
2. Case materials prepared on the basis of research studies undertaken by the faculty.
3. New series of publication entitled "Occasional Papers", has been launched. The papers in this series will critically examine current and arising issues in the area of Research Management, Educational Technology, Administration and Management as well as newer approaches to research planning and prioritisation. The Occasional Paper No:1 entitled "Curriculum Development for Higher Education in Agriculture: Arising Issues" was published.

Besides, the Academy brought out the following periodical publications:

Agricultural Research Management Abstracts (ARMA) is a half yearly publications appearing every June and December. It provides latest information in the specialised areas of Agricultural Research Systems and Policies, Project Management, Human Resources Management, Technological Development, Organisational Development, Technology Transfer, Agricultural Information, Agricultural Education and Agricultural Economics.

NAARM News, published every quarter, brings out the information about the events programmes both present and future and welfare activities of the Academy.

NAARM Alumni News, a half-yearly publication, brings out information for and about the alumni of the Academy.

Green and Glory, a laboratory journal published for every Foundation Course based on the theses of the trainees of a particular batch.

FARM SERVICES UNIT

The campus of the Academy is being developed with an approach to functionality and aesthetics. The landscaping is done with a view to provide a congenial environment for academic activity. Stress is being laid on in utilising a large and varied collection of trees, shrubs, climbers, herbaceous plants and ground covers with an eye on the educational value of a well labelled rich collection of flora.

The Agro-technology Demonstration Blocks have conceived as a show window of the national agricultural research system. The latest technology and crop varieties released by different ICAR institutes and agricultural universities are put on show in one place for the benefit of trainees and visitors comprising national and international scientists and policy makers of agricultural research and development.

Objectives:

Landscaping the campus, developing and maintaining the estate and laying out the agro-technology demonstration blocks in different areas of crops and different silvi systems are the objectives of the unit. The landscaping activities cover entire 50 h.a. of the campus. The Agro-technology Demonstration Blocks cover an area of 30 ha. out of which eight hectares are covered each by field crops and orchards and vegetable cultivation. The remaining area is covered by silvi systems, namely agro-horticulture, silvi-pasture, horti-pasture, agro-forestry, agro-energy systems and plantations like high density plantations, graded height plantations and green belts.

Activities

Field crops:

Fifteen varieties of different field crops like cereals (maize, wheat), millets (sorghum, pearl millet), oilseeds (groundnut, sunflower, niger) were grown during the kharif (1990) and rabi (1990-91) seasons. These include the currently recommended and latest varieties released by ICAR institutes and agricultural universities. Emphasis was given to maximise the productivity with optimal utilisation of available resources.

Horticultural crops:

Demonstration of fruit crops, vegetable crops and flower crops. The fruit crops included several improved varieties and hybrids of guava, sapota, mango, grape,

ber and pomegranate. The mango orchard was expanded by including four more varieties and hybrids like Ratna, Amarpali, H-10 and H-13 released recently by K.K.V.P. (Dapoli) Indian Agricultural Research Institute and Indian Institute of Horticultural Research respectively. Besides the above fruit crops, the table variety of banana -- 'Robusta' was introduced on an area of about a hectare.

Sixteen kinds of vegetables in 40 varieties (transplanted and direct sown) were grown in kharif (1990) and rabi (1990-91) seasons. The Floriculture Block included several varieties of roses and jasmine.

Landscaping:

Importance was given in making the campus green by establishing and maintaining lawns in and around main gathering places like Administrative Block, Academic Block, Faculty Block, Halls of Residence and Scientists Home. Several flowering annuals, borders, hedges and foliage shrubs were added to the existing ones provide aesthetic environment to academic activity.

Silvi Systems:

Besides the conventional input systems for the increased productivity, several systems have been established in order to demonstrate the increase in productivity per unit area of land by way of obtaining food, fuel and fodder from the same piece of land. These include agro-horticultural, agro-forestry, horti-pastoral, silvi-pastoral and energy plantation systems which were developed on low moisture and poor fertility status soils. Fruit species like custardapple, phala, amla and woodapple were included in these systems.

Graded height green belt plantations:

Several useful forest species were established in peripheral and other areas to provide aesthetic environment value to the visitors and trainees in the campus.

Honours and Awards:

The Academy was once again in limelight at the Horticultural and Rose Shows conducted during the year by bagging a number of awards and prizes.

Rose Show: The Academy participated in the Rose Show organised by the Hyderabad Rose Society on December 29, 1990 and bagged 20 prizes including the prestigious 'FACT Shield' for the Best Institutional Rose Garden.

Horticultural Show: Fifteen prizes for the best collection of vegetable and fruit varieties and cut flowers were won at the Horticultural Show organised by the 51st All-India Industrial Exhibition held in February 1991.

ADMINISTRATION AND ACCOUNTS

The Administrative Wing of the Academy is organised into the following four sections to provide effective service:

1. Admn. I Section --- Establishment matters
2. Admn. II Section --- Cash and Bills
3. Admn. III Section --- Stores and Purchase
4. Audit and Accounts --- All financial matters
Section

The Administrative Wing, besides ensuring smooth conduct of various training programmes of the Academy, also provides faculty support in the areas of Administration and Accounts in the training programmes.

The Human Resources

The staff strength rose significantly during the year 1990-91. Five faculty members joined during the period. Similarly eight members joined the other categories of staff.

The category-wise position of human resources during the period is set out in the following Table

Name of the Post/ Category	Total Sanctioned Strength	Posts filled as on 31.03.1991
Scientific	43	21
Technical	41	40
Administrative	49	43
Auxiliary	28	22
Supporting	37	36
	198	162

FINANCIAL OUTLAY

Rupees in hundreds

Rupees in thousands

Sl. No.	Head of Account	Actuals during 1989-90			Revised Estimates 1990-91		
		Non-plan	Plan	Total	Non-plan	Plan	Total
	Pay & Allowances	27,48,000	23,99,600	51,47,600	54,75,000	19,60,000	74,35,000
	Travelling Expenses	88,000	96,000	1,84,000	2,50,000	2,70,000	5,20,000
	Recurring Contingencies	21,24,000	6,88,000	28,12,000	30,75,000	8,00,000	38,75,000
	Non- Recurring Contingencies						
	A) Equipments	-----	9,58,800	9,58,800	-----	28,40,000	28,40,000
	B) Works	-----	33,57,400	33,57,400	-----	21,30,000	21,30,000
	C) Others	-----					
		49,60,000	74,99,800	1,24,59,800	88,00,000	80,00,000	1,68,00,000

PHYSICAL FACILITIES

The Academy has established the following laboratories to make the teaching - learning process more effective, and to help the trainees learn effective use of mass media for dissemination of information.

Video Instruction and Production Laboratory

This laboratory, comprising two television studios and a Master Control Room, facilitates intensive training of scientists in the preparation of effective resource materials for telecasting. It helps in conducting role play exercises for personality development. Video educational programmes are also produced. It is also useful for proper speech communication skills.

Audio-Visual Lab and Photography Unit

The Unit, equipped with all modern photographic and audio reprographic facilities, provides an excellent opportunity to gain experience in the preparation and use of audio-visual aids for effective presentation. These include Overhead Projectors, Epidiascope, Coloured Transparency-making units, Slide projectors, Tape recorders and Public Address Systems.

Speech and Mass Communication Laboratories

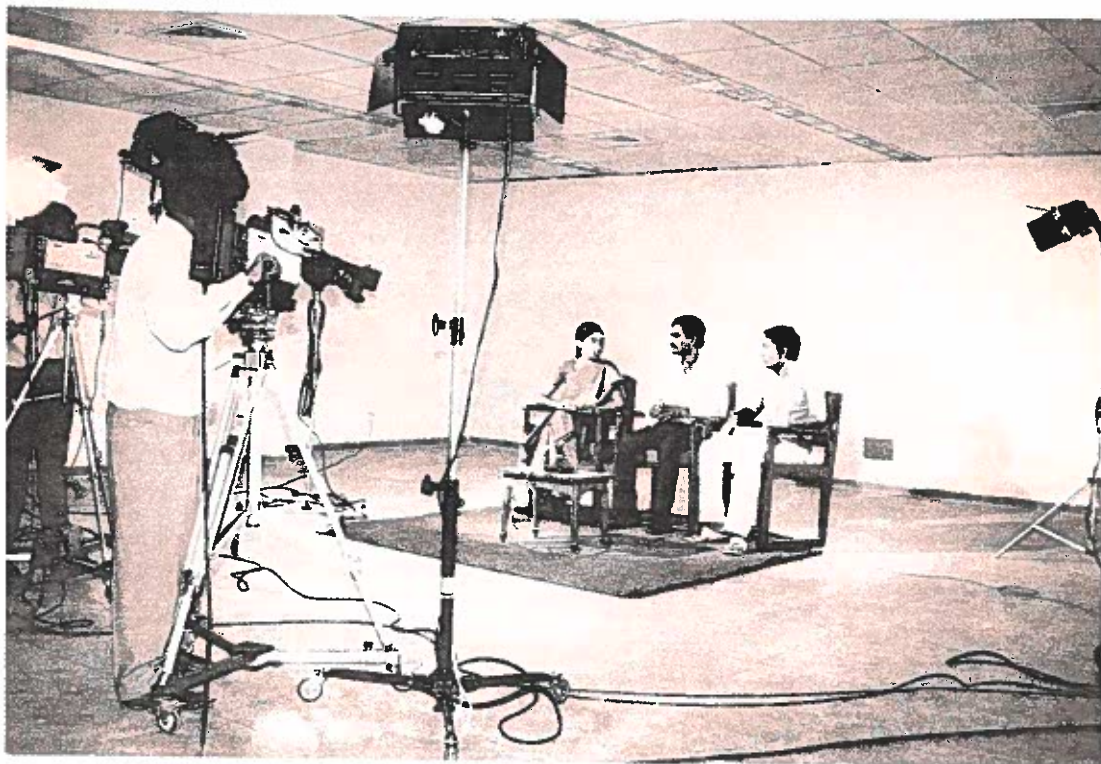
These laboratories facilitate the scientists to record their programmes such as interviews, discussions and talks, specially for the rural listeners.

Organisational Behaviour Laboratories

Individual and group exercises in team-building, leadership and group dynamics conducted in these laboratories provide the trainees an opportunity to understand the various aspects of organisational behaviour including human thought, action and behaviour.

Self-aided Instructional Laboratory

This laboratory provides opportunity for the participants to undergo structured and programmed instruction under self-learning situations. Automated electronic equipments and computerised instruction facilitate the process. A set of carrels with Telex Caramate imparting a tape slide programme, a VCP with monitor for a video tutorial and a computer for a programmed instruction. Software developed for learning Plant Pigments/ Chlorophylls is being evaluated for its effectiveness as a learning tool.



Video Instruction and Production Lab

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Computer Cell

A Computer of mini-frame WIPRO S-306 with 4 MB RAM and 170 MB hard drive along with 12 terminals having 640 K RAM was installed. In addition to this, a DCM TURBO 2000 PC-AT forms a part of the cell. Three dot matrix printers and two Near Letter Quality printers besides one plotter are also acquired. These systems are used to provide intensive training on computers to the trainees of FCARPM, prepares Databases and develop Library Information Service. Besides, computers are also being used in Office Administration for preparation of Paybills, Accounting & Budgeting, Recruitment Processing and secretarial work. The Faculty and Staff are also being trained on the use of various application packages such as Wordstar, dBase and Lotus with a view to increase work efficiency.

Desk Top Publishing System

A Desk Top Publishing System, Apple Macintosh-SE was installed with several software packages including MS Word, Mc Write, Mc Paint, Mc Draw and Hindi Font. These packages would help improve the quality of publications at the Academy.

Banking Facilities

The Central Bank of India, Golconda Fort Branch, Hyderabad, has been operating an Extension Counter (Regular) at the Administrative Block to cater to the banking requirements of the Academy, faculty members, staff and trainees. The counter has now been shifted to the Faculty-cum-Laboratory building. Here the Extension Counter offers safe deposit locker facility to the faculty and the staff members.

Capital Works

The construction of Faculty-cum-Laboratory building was completed and occupied. The construction of the library building is in full progress. The construction of 60 residential quarters is nearing completion. The construction of Irrigation Reservoir has commenced.

DISTINGUISHED VISITORS

A large number of distinguished personalities including national and international scientists visited the Academy during 1990-91. They include:

- ** A three-member Parliamentary Consultative Committee of the Union Ministry of Agriculture, Government of India comprising Dr. Asim Bala, MP, Mr. Chhotubhai Patel, MP and Mr. Channaiah Odeyar, MP.
- ** A six-member group of Asian National Agricultural Research Directors comprising Dr. S.L. Amarasiri, Sri Lanka; Mr. Richard M. Juanillo, Philippines; Dr. S.M. Shah, Nepal; Dr. Sumaro, Indonesia; Mr. U. Aung Thaung, Myanmar and Dr. Gan Xiaosong, China.
- ** A three-member delegation from Somalia Democratic Republic comprising Mr. Bashir Hagi Ahmed Aden, Mr. A.A. Hussain and Ms. Asia Ahmed Aden.
- ** A two-member team from the Philippines with Dr. Dominador D. Ignacio, Vice-President for Research, Extension and Production and Consortium, the Philippines and Dr. Rosita Rose, coordinator, Regional Applied Communication Office (RACO) of Central Luzon University, the Philippines.
- ** Mr. Foruad Elkholy, Director General, Mr. Maudi M. Addel Sanad and Mr. Magdy Roushdy Ishak from Egyptian International Centre for Agriculture, Dokki, Egypt.
- ** A two-member delegation from Indonesia comprising Dr. Engellert Oscar Manuat, Manager, Farming Systems Research, Nusa Tenggara Agricultural Research Project, Kupang and Dr. Abdullah Bamialin.
- ** A six-member agricultural extension team from the Benu State of Nigeria.
- ** A World Bank team comprising Dr. Kidd, Dr. T.C. Jain and Dr. Lodha.
- ** Mr. S.K. Biradar and Dr. M. B. Tippannavar, Member, Board of University of Agricultural Sciences, Dharwar and U.G. Nalawadi, Head, Division of Horticulture, Dharwar.
- ** Dr. Michael Maclean, World Bank Resident Mission, New Delhi.
- ** Dr. Robert Chambers, Administrative Staff College of India, Hyderabad.



Dr.K.V.Raman, Director, addressing the participants of the Workshop on Forestry Faculty Training Project Evaluation.



The Director and Faculty members discussing with the group of Asian Agricultural Research Directors.

- ** Dr. N.S. Randhawa, former Director General, ICAR, New Delhi.
- ** Mr. T.L. Shankar, Director-General, National Institute of Rural Development, Hyderabad.
- ** Dr. Christopher Kalangi, College of Agriculture, University of Arizona, Tuscon, USA.
- ** Dr. John Becker, Office Director USAID.
- ** Dr. Maharaj Singh, Winrock International, New Delhi.
- ** Ms. Anita Frio, Sr. Training Specialist at International Rice Research Institute (IRRI), the Philippines.
- ** Dr. Edward Clay, Director, Relief and Development Institute, London, U.K.
- ** Dr. M.J. Mermillod, FAO, Rome.
- ** Dr. S.P. Ambrose, ICRISAT, Patancheru, Hyderabad.
- ** Mr. Gordon Conwaay, Ford Foundation, New Delhi.
- ** Mr. Vijay S. Pande, Regional Director, IDRC, New Delhi.
- ** Dr. John Farrington, ODI, London.
- ** Mr. S.M.K.B. Nandanaike, ARIA, Colombo, Sri Lanka.
- ** Dr. Zaryhum Woldeyes, Deputy General Manager, Institute of Agricultural Research, Addis Ababa, Ethiopia.
- ** Dr. Steven Anderson, Assistant Professor of Forestry and State Extension Centre, Oklahoma State University, USA.
- ** Dr. Y.L. Nene, Deputy Director General, ICRISAT.
- ** Dr. Charles H. Antholt, World Bank, Washington.
- ** Dr. Collin A. Maclung, India Coordinator, WINROCK International, New Delhi.
- ** Dr. R. Nagarcenkar, Deputy Director General (Education), ICAR, New Delhi.

- ** Dr. Charles R. Hatch, Chief of Party, Forestry Planning Development, Islamabad, Pakistan.
- ** Dr. Douglas P. Richards, Department of Forestry, Mississippi State University, Mississippi State, USA.
- ** Dr. W.E.S. Mutch, Department of Forestry and National Resources, University of Edinburgh, U.K.
- ** Mr. D. R. Chaffey, First Secretary, Natural Resources, British High Commission, New Delhi.
- ** Mr. Suresh Kumar, Director, MANAGE, Rajendranagar, Hyderabad.
- ** Dr. Allen Thomas, Chief, Coverdale Program, MANAGE, Hyderabad.
- ** Mr. Muhammed A. Mannan, Asian Development Bank, Manila.
- ** Dr. R.S. Paroda, Deputy Director General (Crop Sciences) Indian Council of Agricultural Research, New Delhi.
- ** Dr. I.P. Abrol Deputy Director General (Soils, Agronomy and Engineering) ICAR, New Delhi.
- ** Dr. P.V. Dehadrai, Deputy Director General (Fisheries) ICAR, New Delhi.
- ** Dr. C.Prasad, Deputy Director General (Transfer of Technology and Agricultural Extension), ICAR, New Delhi.
- ** Dr.K.L. Chadha, Deputy Director General (Horticulture), ICAR, New Delhi.
- ** Dr. R.M. Acharya, Deputy Director General (Animal Sciences), ICAR, New Delhi.
- ** Dr. T.P. Ojha, Deputy Director General (Engineering), ICAR, New Delhi.
- ** Dr. N.G.P. Rao, Chairman, ASRB, New Delhi.
- ** Dr. D.K. Das Gupta, Vice Chancellor, Bidhan Chandra Krishi Vishwa Vidyalaya, Mohanpur Nadia, West Bengal.
- ** Dr.D.K. Sharma, Vice Chancellor, J.N. Krishi Vishwa Vidyalaya, Jabalpur.

- ** Dr. S.B. Kadrekar, Vice Chancellor, Marathwada Agricultural University Parbhani, Maharashtra.
- ** Dr. P.C. Bora, Vice Chancellor, Assam Agricultural University, Jorhat.
- ** Dr.A.L. Chaudhry, Vice Chancellor, Haryana Agricultural University, Hissar.
- ** Dr. V.P. Shukla, Vice Chancellor, Indira Gandhi Krishi Vishwa Vidyalaya, Krishi Nagar, Raipur.
- ** Dr. K.S. Gill, Vice Chancellor, Punjab Agricultural University, Ludhiana.
- ** Dr. R.K. Thakur, Vice Chancellor, Y.S. Parmar University of Horticulture and Forestry, Solan, H.P.

PUBLICATIONS

Books

1. Samanta R.K. (1990), Rural Development in North-East India: Perspective, Issues and Experiences, Uppal Publishing House, New Delhi.

Scientific / Review Papers

1. Raman K.V. and Balaguru T. (1990). National Agricultural Research Systems in India - Journal of Indian School of Political Economy 2 (3): 449-474, 1990.
2. Raman K.V. and Balaguru T. (1990). Human Resources Development as a key to the success of Farming System Research in India. Journal of Farming System Research - Extension 1 (1). 153-164, 1990.
3. Sudarsanam R. (1990). A Form for Popular Communication for Development, and Anticipatory Management, Vigyan and Pradyogik Samachar, A.K. Prachart Noumana Grameen Vikas, NIRD, Hyderabad July-December 1990.
4. Sudarsanam R. (1990). Ergonomics in Curriculum for Agricultural Engineering, Journal Ergonomics, Bhopal September 1990.
5. Samanta R.K. (1990). The Political Economy of Panchayat in West Bengal, Social Change 20 (1).
6. Samanta R.K. and Chelvan R.C. (1990). Development Communication - Key to Eradicate Rural Poverty. Khadi Gramodyog 36 (4).
7. Samanta R.K. (1990). Rural Development - Needs a Balanced and Integrated Approach, Khadi Gramodyog 36 (2).
8. Samanta R.K., Chelvan R.C., Gadewar A.U. (1990). Users' Participation in Technology Generation- Key for Effective Technology Transfer in Agriculture and Rural Development, D. Das Gupta (Ed.), College of Agriculture, Vishwa Bharati, West Bengal (accepted).
9. Sabarathnam VE (1989). Selection of Psychological Factors as Independent Variables. Journal of Research APAU XVII (2) pp. 159-163.
10. Sabarathnam VE (1990). Research Climate and its Impact on Efficiency of Scientists in Indian Dryland Research System. Indian Journal of Applied Psychology Vol.2-7 (2) pp 84-91.

MISCELLANIA

QUAMI EKTA WEEK

A week-long celebrations were held during the Quami Ekta Week at the Academy from November 19 to 25, 1990.

The celebrations began with the Academy's staff, faculty and participants of various training programmes taking national integration pledge on the first day.

Book exhibitions were organised during the celebrations on themes like welfare of minorities, weaker sections, and environment and cultural conservation.

Debate, elocution competitions and cultural programmes on these themes were organised by the participants of the 35th FCARPM.

Dr. V. Raghavendra Rao, Principal Scientist, coordinated the celebrations.

HINDI DAY

The Academy celebrated Hindi Day on September 14, 1990 under the Chairmanship of Dr. K.V. Raman, Director, NAARM. Dr. Krishna Kumar Goswamy, Professor, Post Graduate Centre, Dakshin Bharat Prachar Sabha, Hyderabad was chief guest.

Dr. Raman in his presidential address stressed the need for single official language for a country like India with multi-linguistic character. He advised the staff to learn Hindi as a link language for the existence and to be in touch with the people of other parts of the country.

Dr. Krishna Kumar Goswamy, Professor, PG Centre, Dakshin Bharat Prachar Sabha, Hyderabad, advised the people to give a thought to national language and erase the misgivings that it is the language of a particular region. He cautioned that in the beginning there will be some difficulty particularly in the use of science terminology in Hindi. But once it gets started there will be a smooth sailing.

Dr. Devesh Kishore, Officer-in-charge, Hindi explained the various activities and achievements of the Hindi Unit. Mr. D. Venkateswarlu, Jr. Hindi Translator welcomed the gathering.

LITTER-FREE CAMPUS

The Academy observed Litter-Free Campus week during the first week of the New Year 1991. The week was observed with a view to inculcating a sense of commitment among the residents of campus, the staff and participants of the various courses to keep the campus clean and healthy.

Works related to clearance of weeds and garbage, were taken up during the week. Further, posters with slogans emphasising the need to keep the campus clean were put up at important places in the campus.

FACULTY DEVELOPMENT

Participation and Presentation of Papers

The Faculty participated in various conferences/ symposia/ workshops and meetings held during 1990-91 where in they presented their paper and exchanged views with the delegates.

The details are as follows:

1. Dr. K.V. Raman, presented a paper on Curriculum Development for Higher Education in Agriculture: Arising Issues for 21st Century in the National Seminar on Agricultural Education on February 3, 1990 at Haryana Agricultural University, Hissar.
2. Dr. Jagdeesh C. Kalla submitted a Case Study on Economic Viability of Goat Farming under Intensive Management System in International Seminar for Goat Production held at Makdoom, UP.
3. Dr. Devesh Kishore presented a paper entitled Strategy for Social Forestry Extension - Present Status and Needed Changes held from October 25-29, 1990, at University of Agricultural Sciences, Bangalore.
4. Dr. R. Sudarsanam presented a paper on Curricular Issues for Agro-forestry in the Workshop on Forestry Faculty Training Sub-Project Evaluation conducted by ICAR at NAARM held from December 11-13, 1990.
5. Dr. S.N. Saha presented a paper on Employment Opportunities for Forestry and Agro-forestry graduates in workshop on Forestry Faculty Training-Evaluation held from December 11-13, 1990 at NAARM, Hyderabad.
6. Dr. R. Sudarsanam presented a paper on Social Sciences Research: Types and Features with Special Reference to Safe Systems Approach at UNDP supported Workshop organised by Home Science College, APAU, Hyderabad on February 15, 1990.
7. Dr. R. Sudarsanam presented a paper on Anticipatory Research Management and Participatory Process in the Workshop on Futurology and Rural Development conducted by NIRD, Hyderabad on January 7-12, 1991.

8. Dr. VE. Sabarathnam presented a paper on 'Scale Development' in the workshop on Field Research Methods organised by the College of Home Science, APAU, Hyderabad from February 13-16, 1991.
9. Dr. T. Balaguru presented a paper on Experimental Research Methods in the Workshop on Field Research Methods organised by the College of Home Sciences, APAU, Hyderabad from February 13-16, 1991.
10. Dr. T. Balaguru presented a paper on Training of Trainers in Interdisciplinary Seminar on Human Resources Development organised by the Central Institute of English and Foreign Languages, Hyderabad from January 19-20, 1991.

Honours

Mr. A.U. Gadewar, Associate Professor, was awarded the Ph.D degree in Agriculture Extension by the Indian Agricultural Research Institute, New Delhi, on February 8, 1991 for his thesis on "Management of Training and Visit System in Maharashtra -- An Empirical Study".

FOREIGN VISITS

- * Dr. K.V. Raman, Director, NAARM, visited Dhaka, Bangladesh to attend the Governing Board Meeting of the SAARC Agricultural Information Centre (SAIC) on May 14-15, 1990 for reviewing and planning the activities of the Centre. He also visited the Agricultural University Campus at Mymensing and had discussions with some of their professional staff.
- * Dr. K.V. Raman, Director, NAARM, visited Australia from October 31-November 14, 1990 to finalise the report of the project on "Evaluation of use and impact of the Sorghum". A collaborative study with the University of Queensland, Brisbane, Australia, supported by the Australian Council International Agricultural Research, on the impact of their publication "Nutritional Disorders in Sorghum". NAARM conducted field study, analysis of responses and data and preparation of the Indian section of the report.
- * Dr. K.V. Raman, Director, NAARM, visited Philippines from November 19-25, 1990 to participate in the workshop on "Management Information System". The first part of the workshop was conducted at NAARM. Dr. Raman also held discussions with Director, Southeast Asian Regional Centre for Graduate Study and Research in Agriculture (SEARCA), about the various programmes and activities of SEARCA, and also with the Senior Officers of the Asian Development Bank.

- * Dr. Jagdeesh C. Kalla, Joint Director, NAARM, visited Philippines from November 12-23, 1990 to attend International Workshop on "Management Information Systems for Agricultural Research" organised by the International Services for National Agricultural Research (ISNAR). The Workshop was organised to develop and evaluate new approaches organising and using good information for Research Management.

TRAINING

- * Dr. K.P.C.Rao, Associate Professor, NAARM, underwent advanced training in Agricultural Research and Education Management at Economic Growth Centre, Yale University, New Haven, USA from February 8, 1990 to June 22, 1990 under the supervision of Professor Robert E. Evenson, with special emphasis on "Estimating rates of return on the investments made in Agricultural Research in India".
- * Mr. R. Chittiraichelvan, Associate Professor, NAARM, underwent advanced training in the USA from May 31 to October 12, 1990. He participated in four courses - Video Communication I, Video Communication II, Communication for Social Change and Participatory Approach to Action Research at the Department of Communication, College of Agriculture and Life Sciences, Cornell University, Basic Workshop on Videodisk Design at University of Nebraska and a visit to Interactive Telecommunication Department of New York University. Hands-on experience was obtained in the use of small format video for development and instruction, participatory approaches to development, and interactive multimedia production. Several video-tapes were produced during the training. The proposal enumerating various activities for next five years to augment the video facilities of NAARM and SAUs was prepared in collaboration with Dr. Shirley A. White, Prof. of Communication at Cornell University.
- * Dr. A. U. Gadewar, Associate Professor, attended advanced training in USA from September 3, 1990, to January 11, 1990. He participated in INTERPAKS course on "Organisation and Management of Agricultural Extension Systems: A New Look at Knowledge Transfer" held at University of Illinois, Urbana-Champaign from September 3-28, 1990. He underwent a fall quarter at Ohio State University, Columbus and participated in the courses of i) Research Methods; ii) Research Designs; iii) Methods of Teaching Agriculture; iv) Instructional Media in Agriculture and v) Third World Development. In addition, he participated in professional workshops/seminars and interacted with the faculty of Extension Education Department and cooperative Extension Service of University of Wisconsin, Madison, on the aspects of Research Extension Model, and Prioritisation of Extension Programmes.

- * Dr.K.M. Reddy, Associate Professor, NAARM, underwent advanced training in USA for a period of four-and-a-half months from September 20, 1990. He audited number of Graduate Courses in the areas of Computer Assisted Instruction, Instructional Technology and Educational Administration at the University of Oregon, during fall, and had an exposure to NovaNet System at Institutions in Eugene, Oregon and interacted with Educators. He also discussed with teachers and researchers regarding the appropriate use of computers in their departmental instructional schemes and Research at the University of Illinois, Urbana-Champaign.
- * Dr. A. Gopalam, Associate Professor, NAARM, left for USA on March 19, 1991, for advanced study in the field of educational methodology and instructional technology for a period of four-and-a-half months at Michigan State University under the UNDP Project on Advanced Centre for postgraduate Agricultural Research and Education.

NAARM ALUMINI ASSOCIATION

The total membership of the Association stands at 473 as on March 31, 1991.

Following are the office-bearers of the Association for 1991-92:

PATRON	:	Kapila, Director General, ICAR
PRESIDENT	:	Dr. K.V. Raman, Director, NAARM
GENERAL SECRETARY	:	R. Chittiraichelvan, Scientist (SG), NAARM
VICE PRESIDENTS (2)		
Local	:	Dr. Vilas A. Topani, Scientist NRCS, Hyderabad.
All India	:	Dr. A.K. Saxena, Scientist CIAE, Bhopal
JOINT SECRETARIES (5)		
Eastern Region	:	Dr. K.K. Sharma ILRI, Ranchi
Northern Region	:	Dr. Sanjeev Gupta, Scientist, IGFRI, Jhansi
Western Region	:	A. Chaturvedi NSSB LUP, Nagpur
Central Region	:	U.R. Badegaonkar CIAE, Bhopal
Treasurer	:	Sh. C.P. Mohan Kumar Finance & Accounts Officer NAARM, Hyderabad.

IMPORTANT COMMITTEES

Management Committee

The Academy is guided in planning, organising and conducting various types of training programmes and research studies by the Management Committee of NAARM. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation both in terms of physical targets and schedules and provide broad guidelines for future programmes. The Management Committee also helps in formulation and consideration of proposals for Five Year Plan and Annual Plan and takes up periodical review of progress of development schemes.

The Management Committee, besides assisting Director of the academy in the discharge of his functions, concentrates his attention on the research and other programmes of the academy and ensures their implementation. While examining these programmes periodically, it pinpoints bottlenecks, if any, and suggests remedial measures.

The 16th and 17th meetings of the Management Committee of the Academy were held on July 21, 1990 and January 19, 1991 respectively. The composition of the Management Committee of NAARM is as follows:

- | | | |
|----|--|----------|
| 1. | Dr. K.V. Raman
Director, NAARM
Rajendranagar
HYDERABAD - 500 030 | Chairman |
| 2. | Dr. Jagdeesh C. Kalla
Joint Director, NAARM
Rajendranagar
HYDERABAD - 500 030 | Member |
| 3. | Dr. N.K. Bhattacharya, Director
Central Institute for
Research on Goats
P.O. FARAH 281 122
MATHURA (U.P) | Member |
| 4. | Dr. V.R.P. Sinha, Director
Central Institute of
Fisheries Education
Versova, BOMBAY 400 058 | Member |

- | | | |
|-----|---|------------------|
| 5. | Dr. N.T. Singh, Director
Central Soil Salinity
Research Institute
KARNAL 132 001 (Haryana) | Member |
| 6. | Mr. V.S. Sampath
The Commissioner &
Director of Agriculture
HYDERABAD | Member |
| 7. | The Director of Animal Husbandry
Govt. of Uttar Pradesh
LUCKNOW (U.P) | Member |
| 8. | Dr. A. Appa Rao
Vice-Chancellor
Andhra Pradesh Agricultural University
Rajendranagar
HYDERABAD - 500 030 | Member |
| 9. | Mr. N.B. Reddy
Post: Aurad
(Sirsi), Dist. Bidar
KARNATAKA | Member |
| 10. | Dr. D. Bap Reddy
33/4 RT, Barkatpura
HYDERABAD 500 027 | Member |
| 11. | Dr. R. Nagarcenkar
Deputy Director General (Edu.)
Indian Council of Agril. Research
Krishi Bhavan
NEW DELHI 110 001 | Member |
| 12. | Mr. D.K. Srivastava
Director (Finance)
Indian Council of Agril. Research
Krishi Bhavan
NEW DELHI 110 001 | Member |
| 13. | Mr. S. Rengan
Chief Admn. Officer
NAARM, Rajendranagar
HYDERABAD 500 030 | Member-Secretary |

Steering Committee on Advanced Centre on Agricultural Education and Research Management

This Committee was initially called the Sub-Project Advisory Committee and has been later renamed as Steering Committee. The 6th and 7th meetings of the Steering Committee on the Advanced Centre on Agricultural Education and Research Management were held on July 14, 1990 and January 18, 1991 respectively, at NAARM, Hyderabad. The Committee reviewed the progress of the project and provided broad guidelines for its implementation based on the suggestions of user agencies. The following are the members:

- | | | |
|----|---|----------|
| 1. | Dr. K.V. Raman
Director, NAARM
Rajendranagar
HYDERABAD - 500 030 | Chairman |
| 2. | Dr. R. Nagarcenkar
Deputy Director General (Edu.)
Indian Council of Agril. Research
Krishi Bhavan
NEW DELHI - 110 012 | Member |
| 3. | Dr. Upadhyay
Vice-Chancellor
Narendradeva University of
Agriculture & Technology
FAIZABAD - 224 229 (UP) | Member |
| 4. | Dr. S. Jayaraj
Vice-Chancellor
TNAU, COIMBATORE - 641 003 | Member |
| 5. | Dr. A. Appa Rao
Vice-Chancellor
APAU, Rajendranagar
HYDERABAD - 500 030 | Member |
| 6. | Dr. R.C. Mehta
Dean (Agril.)
Rajasthan College of Agriculture
RAU, UDAIPUR - 313 001 | Member |

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|-----|--|------------------|
| 7. | Dr. O.N. Kunzru
Jt. Director (Extn. Edu.)
IVRI, IZATNAGAR - 243 122 (UP) | Member |
| 8. | Dr. P. Geervani
Dean of Home Science
APAU
Rajendranagar
HYDERABAD - 500 030 | Member |
| 9. | Dr. Jagdeesh C. Kalla
Joint Director, NAARM
Rajendranagar
HYDERABAD - 500 030 | Member |
| 10. | Dr. Devesh Kishore
Principal Scientist
NAARM, Rajendranagar
HYDERABAD - 500 030 | Member |
| 11. | Dr. V. Raghavendra Rao
Principal Scientist
NAARM, Rajendranagar
HYDERABAD - 500 030 | Member |
| 12. | Dr. Maharaj Singh
WINROCK International
P.O. 8808, T. Poorvi Marg
Vasant Vihar
NEW DELHI - 110 021 | Member |
| 13. | Dr. George H. Axinn
FAO Rep. in India & Bhutan
FAO of the United Nations
55 Max Mueller Marg
(Ex. Lodi Estates)
NEW DELHI - 110 001 | Member |
| 14. | Mr. S. Rengan
Chief Admn. Officer
NAARM, Rajendranagar
HYDERABAD - 500 030 | Member-Secretary |

Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provided periodical suggestions for implementation of official Language policy and rules and regulations of Govt. of India that are issued from time to time. The committee met on February 26, 1991 and discussed several issues relating to the implementation of Hindi language. The following are the members:

- | | | |
|-----|--|------------------|
| 1. | Devesh Kishore
Principal Scientist | Chairman |
| 2. | Dr. K. Venkateswarlu
Scientist (SG) | Member |
| 3. | Dr. A. Gopalam
Scientist (SG) | Member |
| 4. | Dr. Jagannatham Challa
Scientist (SG) | Member |
| 5. | Mr. R.B. Gaddagimath
Librarian | Member |
| 6. | Mohd. Habeebuddin
Chief Admn. Officer i/c | Member |
| 7. | Mr. P. Mohan Kumar
Finance & Accounts Officer | Member |
| 8. | Mr. M. Srinath Reddy
Public Relations Officer | Member |
| 9. | Mr. R.V.V.S. Prakasa Rao
Editor | Member |
| 10. | Mr. P.P. Brahmaji
Assistant | Member |
| 11. | P. Swamy
SSG-I | Member |
| 12. | Mr. D. Venkateswarlu
Jr. Hindi Translator | Member-Secretary |

PERSONNEL

K.V. Raman

Director

Director's Personal Cell

L. Jhansi Lakshmi
T. Srinivas
T.V. Rama Das
P.Swamy

Sr. Stenographer
Sr. Clerk
Jr. Stenographer
SSG-I

AGRICULTURAL RESEARCH SYSTEMS & POLICIES UNIT

Jagdeesh C. Kalla
P. Kumar
VE. Sabarathnam
T. Balaguru
K.P.C. Rao
K. Vidyasagar Rao
Sarada Samanta
A. Vijaya Lakshmi

Joint Director
Principal Scientist
Scientist SG
Scientist SG
Scientist SG
Scientist SG
Sr. Stenographer
Jr. Stenographer

(A) Computers

U. Muralidhar
M.J. Pradeep Kumar
K. Venkateswara Kumar
C.V.V.S.S.R.L. Phani Raj

Programmer (Computer)
Computer Operator
Statistical Assistant
Jr. Clerk

EDUCATIONAL SYSTEMS AND TECHNOLOGY UNIT

Devesh Kishore
R. Sudarsanam
K.M. Reddy
A. Gopalam
K.R. Prabhakar
N.R. Nageswara Rao
Ch. Janardhan Rao
P. Anantha Reddy
N. Naresh Kumar
M. Venkatesh

Principal Scientist
Principal Scientist
Scientist SG
Scientist SG
Programme Officer (CCTV)
A.V. Operator
Video Cameraman
Technician Instruments
VCR-cum-Tele Cine Operator
Stenographer

(a). Hindi Section

D. Venkateswarlu
J. Renuka
S. Pradeep Singh
K. Srinivasa Rao

Jr. Hindi Translator
Hindi Assistant
Hindi Translator
Jr. Clerk

PROJECT MANAGEMENT UNIT

V. Raghavendra Rao
K. Venkateswarlu
S. N. Saha
P. Manikandan
T. Vanisri

Principal Scientist
Scientist SG
Scientist SG
Scientist SG
Jr. Stenographer

TRANSFER OF TECHNOLOGY UNIT

R.K. Samanta
R.C. Chelva
A.U. Gadewar
L. Venkateswarlu
Bansidhar Nayak
D. Rajagopala Rao
S. Sunder Raj
P. Anand Kumar

Scientist SG
Scientist SG
Scientist SG
Photographer
Projectionist
PAE Operator
Media operator
Jr. Stenographer

HUMAN RESOURCES DEVELOPMENT UNIT

M.M. Anwer
Jagannadham Challa
S. Shanmugam
Y. Anuradha

Scientist SG
Scientist SG
Scientist SG
Jr. Stenographer

INFORMATION AND DOCUMENTATION UNIT

R.B. Gaddagimath
P.V. Nirmala
Y.V. Siva Prasad
T. Shanmuganahan
A. Mercy
D. Dhanalakshmi
Venkatesham
M. Narsing Rao

Librarian
Cataloguer
Documentation Assistant
Library Assistant
Jr. Stenographer
Jr. Clerk
Xerox Machine Operator
SSG-I

PRODUCTION AND PUBLICATION UNIT

R.V.V.S. Prakasa Rao
N. Narasimha Reddy
A. Murali Mohan Rao
K. Anil Kumar
P. Namdev
Y. Vijay Kumar
M. Shekhar Reddy
S. Swamy
Prashant Gaikwad
C. Chandramouli

Editor
Off-set Machine Operator
Artist
Reprographic Assistant
Graphic-cum Visualiser
Assistant Editor
Darkroom Assistant
Asst. Gestt. Operator
Binder
SSG-I

FARM SECTION

V. Murali
K. Rangasamy
M.A. Basith
P. Vijender Reddy
B. Veeraiah
Ahire Laxman Maharu
K. Prabhudas
M. Padmaiah
K. Veeranarasaiah
P. Eswari
P. Balraj
M. Yadaiah
K. Satyanarayana
N. Sukunamma
T. Jangamma
S. Jangaiah
B. Bharatamma
J. Chandraiah
G. Pentaiah
R. Sattaiah
G. Anasuya
K. Hemlal

Farm Superintendent
Field Assistant
Lab Technician
Museum Assistant
Farm Assistant
Farm Assistant
Sr. Clerk
Tractor Driver
Carpenter-cum-Painter
Power Tiller Driver
SSG-I
SSG-I
SSG-I
SSG-I
SSG-I
SSG-I
SSG-I
SSG-I
SSG-I
SSG-I

HOSTEL SERVICES

Zameer Ahmed
Sham Bahadur
P. Neelakantam
D. Sangamma

Manager (Hostel Services)
Catering-in-charge
Sr. Clerk
Cook

G.V. Bikshapathi
 B. Santamma
 B. Subba Rao
 Shakunthla
 G. Dasaradha
 A. Yadaiah
 M. Krishnaiah
 C. Anjaiah
 K. Kalavathi
 M. Yadamma
 I. Bharatamma

SSG-I
 SSG-I
 SSG-I
 SSG-I
 SSG-I
 SSG-I
 SSG-I
 SSG-I
 SSG-I
 SSG-I
 SSG-I

ADMINISTRATION

S. Rengan
 Mohd. Habeebuddin
 M. Srinath Reddy
 K. Janakiramaiah
 T. Ashok Kumar
 Y. Sankara Rao
 P.P. Brahmaji
 K.S. Sharma
 N. Raghunath
 P.G. Kohad
 W. Srinivasa Bhat
 M. Narasimha Rao
 P.P. Sastry
 M. Dinesh
 K. Swarajya Lakshmi
 G. Raj Reddy
 T. Prasanna Kumar
 T. Pardhasaradhi
 B. Padma Saroja
 P. Srinivasu
 P.R.R. Jyothi
 P.D. Jessie Ecclicia
 M. Sridhar
 C.J. Samuel
 C. Bikshapathi
 M. Sham Rao

Chief Administrative Officer
 Asst. Admn. Officer
 Public Relations Officer
 Superintendent
 Assistant
 Assistant
 Assistant
 Assistant
 Stenographer
 Assistant
 Sr. Clerk
 Sr. Clerk
 Jr. Stenographer
 Jr. Clerk
 Jr. Lib. Asst
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 PMBX Operator-cum-Receptionist
 Jr. Clerk
 Jr. Clerk
 SSG-I
 SSG-I

(a) Vehicles

K. Sattaiah	Driver
S.K. Peera	Driver
M.M. Mohan Rao	Vehicle Mechanic
Mohd. Naseer Khan	Driver
G. Muthyalu	Driver
U. Venkataratnam	Driver
N. Ashok	Driver
B. Narayana	SSG-II

(b) Security

B. Ch. Satyanarayana	S.O.-cum-Caretaker
N. Prabhakara Rao	Plumber
G. Mani Bai	SSG-II
B.H. Dharma Raj	SSG-II
Phool Kumar	SSG-I
Khalid	SSG-I
M. Ashok	SSG-I
L. Satyanarayana	SSG-I
M. Eswaramma	SSG-I

(c) Electrical

M.V.V.S. Ravindra Babu	Electrical Overseer
L. Ramesh	Electrician-cum-plumber
M.K. Sonkusare	Electrician
M. Mohan Rao	Technician (Electrical)
M.K. Shamsuddin	Technician (Electrical)
B.S.N. Murthy	Technician (Electrical)
K. Shivaiah	Technician (Electrical)
K. Obulpathi	Technician (Electrical)
B. Satyanarayana	Technician (Electrical)
A. Nageswara Rao	Lineman
D.R. Sahadev Rao	Pump Driver

AUDIT & ACCOUNTS SECTION

C.P. Mohan Kumar	Finance & Accounts Officer
C. Bagaiah	Assistant
M. Girija Kumari	Jr. Stenographer
S. Balakamesh	Jr. Clerk
R. Chandra Babu	Jr. Clerk

APPOINTMENTS

Following appointments were made during the period:

S.No:	Name of the employee	Designation
1.	T. Vanisri	Junior Stenographer
2.	G. Anasuya	SSG-I
3.	K. Shivaiah	Technician (Electricial)
4.	K. Hemlal	SSG-I
5.	S. Pradeep Singh	Hindi Translator
6.	P. Namdev	Graphic-cum-Visualiser
7.	P. Vijender Reddy	Museum Assistant
8.	S. Rengan	Chief Administrative Officer
9.	U. Muralidhar	Programmer (Computer)
10.	K.R. Prabhakar	Programme Officer (CCTV)
11.	M. Sridhar	Jr. Clerk
12.	R. Chandra Babu	Jr. Clerk
13.	T.V. Rama Dass	Jr. Stenographer
14.	C. Julius Samuel	Jr. Clerk
15.	P.P. Sastry	Jr. Stenographer
16.	P. Anantha Reddy	Technician (Instrumentation)
17.	Y. Anuradha	Jr. Stenographer
18.	B. Satyanarayana	Technician (Electricial)
19.	Sunder Raj	Media Operator
20.	Y. Vijay Kumar	Assistant Editor

Resignations

The following employees of the Academy were relieved from the service to take up new assignment at other organisations during the period:

1.	V. Govinda Raju	Jr. Clerk
2.	Y. Venkateswara Rao	Technician (Electricial)
3.	Dr. K. Pratap Reddy	Scientist (SG)

Promotions

The following members of the staff were promoted on recommendation of the Assessment Committee/Departmental Promotion Committee during the period:

S.No.	Name of the employee	Designation	Promoted
1.	V. Murali	T-5	T-6
2.	N.R. Nageswara Rao	T-4	T-5
3.	A. Murali Mohan Rao	T-II-3	T-4
4.	K. Rangasamy	T-II-3	T-4
5.	M.A. Basith	T-II-3	T-4
6.	B.Veeraiah	T-2	T-I-3
7.	R.V.V.S. Prakasa Rao	T-4	T-6
8.	D. Venkateshwarlu	T-4	T-5
9.	C.Bagaiah	Sr. Clerk	Assistant
10.	P.G. Kohad	Sr. Clerk	Assistant
11.	N. Raghunath	Jr. Steno	Steno
12.	M. Venkatesh	Jr. Steno	Steno
13.	T. Srinivas	Jr. Clerk	Sr. Clerk
14.	K. Prabhudas	Jr. Clerk	Sr. Clerk

Merit Increments

The following members were granted advance increments on the recommendation of the Assessment Committee:

1.	L.Venkateshwarlu	T-II-3	One merit increment
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Transfers

The following transfers were made during the period (NAARM to other ICAR institutes and vice-versa):

S.No	Name of the employee	Designation	From	To
1.	Dr. S.Shanmugam	Scientists SG	CMFRI, Veraval	NAARM, Hyderabad
2.	K. Surekha	Jr.Clerk	NAARM, Hyderabad	CRIDA, Hyderabad
3.	T.Prasanna Kumar	Jr. Clerk	NDRI SRS, Bangalore	NAARM, Hyderabad

Retirements

Dr. K.V. Subba Rao, Principal Scientist, retired from NAARM/ICAR services

Deputations

The following members of the staff were sent back to their parent institutes on expiry of deputation period:

K.Satya Priya	Assistant	DRR, Hyderabad
G.Narasimha Reddy	Superintendent	CTRI, Rajahmundry



National Academy of Agricultural
Research Management
Rajendranagar, Hyderabad 500030 (A.P.)